

MEMORANDUM
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DAPCA: A COMPUTER PROGRAM FOR
DETERMINING AIRCRAFT DEVELOPMENT
AND PRODUCTION COSTS

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SUMMARY

This Memorandum presents a description of a computer program (DAPCA) developed by the author to compute development and production costs for the major subsystems (i.e., airframes, engines, and avionics) of flyaway aircraft. This program is based on the methodologies described in RM-4585-PR, RM-4670-PR (abridged), and RM-4851-PR (abridged).*

Appendices listing the equations used in the program as well as the program itself are also included.

* See footnotes in Section I.

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I. INTRODUCTION

The DAPCA computer program described in this Memorandum computes development and production costs for the major subsystems of flyaway aircraft (i.e., airframes, engines, and avionics).^{*} The cost outputs generated by this program are in the form of cost-quantity unit and cumulative-average improvement curves.

A maximum of 27 inputs are used in the program, of which no more than 12 to 15 are usually required to be entered for a particular run. Most of the major inputs relate to aircraft and engine performance characteristics, such as gross takeoff weight, speed, engine type, engine thrust, etc.

This program was developed primarily in support of the studies reported in RM-4845-PR, RM-4670-PR (abridged), and RM-4851-PR (abridged).^{**} For this reason, a detailed discussion of the basic methodologies underlying the program is not given, since such a presentation would duplicate much of that given in the above-mentioned references. Rather, the presentation in this Memorandum is oriented toward a general description of the computer program, program considerations, and program outputs. For those readers who may be interested in the specific details of the program, all equations used in the calculations are listed in Appendix A. In addition, flow charts of the program are shown in Appendix B, and a listing of the FORTRAN-IV source program is presented in Appendix C.

^{*}The avionics development cost is not computed in the program, but is treated as a thruput. ("Thruput" is the term given to any data input that is entered in the program essentially as a final estimate. In contrast to a data input that is used by the program in its calculations, a thruput goes through the program without processing, to the stage at which the final tabulations and data summaries are produced.)

The term DAPCA is taken from the words Development and Production Costs for Aircraft.

^{**}G. S. Levenson and S. M. Barro, Cost Estimating Relationships for Aircraft Airframes, The RAND Corporation, RM-4845-PR, December 1965 (Privileged Information); A. Frank Watts, Aircraft Turbine Engines--Development and Procurement Cost, The RAND Corporation, RM-4670-PR (abridged), November 1965; C. Teng, An Estimating Relationship for Fighter/Interceptor Avionic System Procurement Cost, The RAND Corporation, RM-4851-PR (abridged), May 1966.

II. GENERAL DESCRIPTION

FLOW OF OPERATIONS

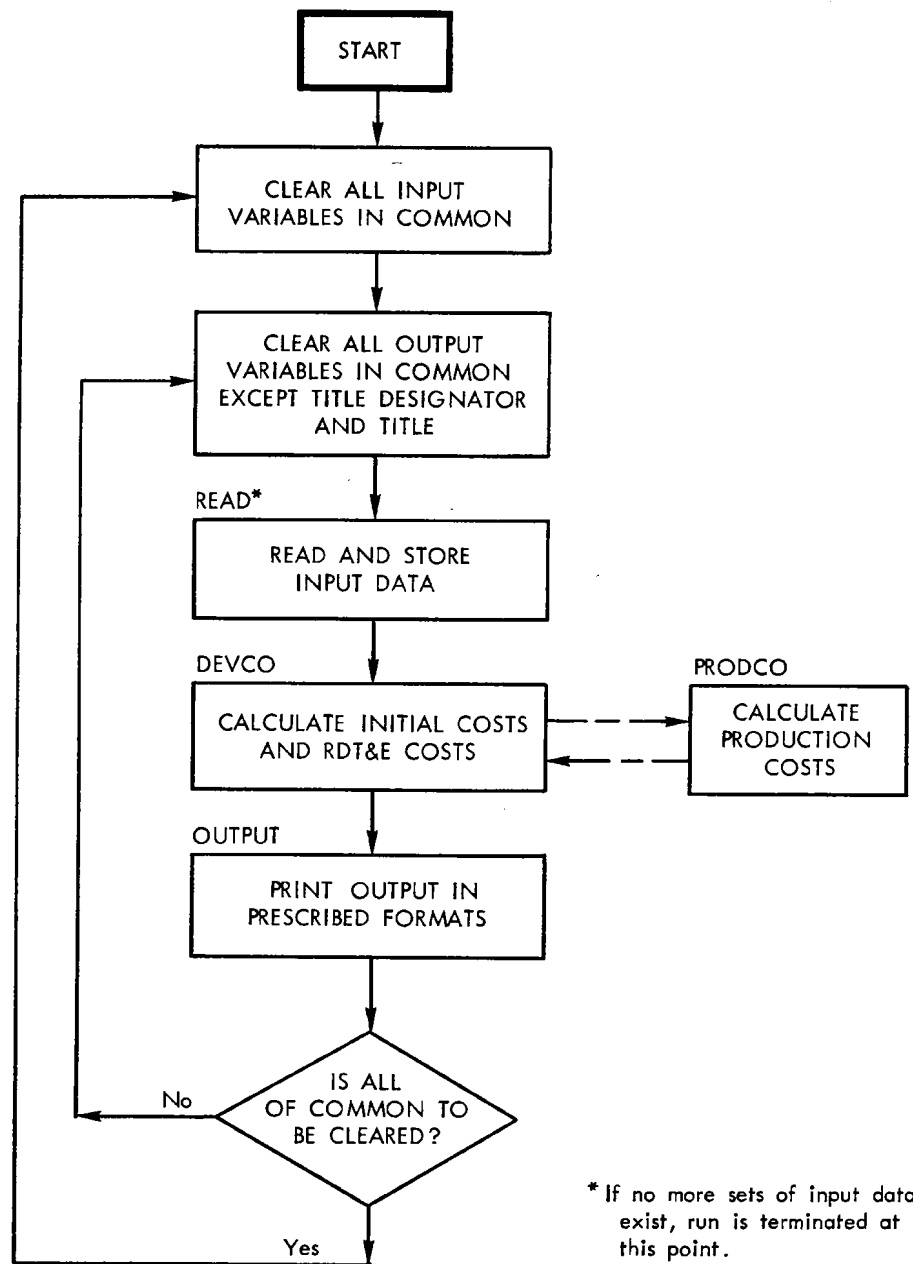
As many runs as desired may be made using the program, with each run requiring approximately three to four seconds execution time. Briefly, the program operates as follows (see Fig. 1 and Appendices B and C). The MAIN routine clears all storage locations in the COMMON region and then calls the various subroutines in the following order. First, subroutine READ is used to read in the data and store the values of the input variables according to their relative locations in COMMON. This subroutine also makes certain checks on, and corrections to, the data (if required). Subroutine DEVCO is called upon next to compute the constant portions (independent of aircraft quantity) of the production or recurring cost equations. This subroutine then uses subroutine PRODCO to calculate the airframe, engine, and avionics production costs. Subroutine DEVCO then uses those results from subroutine PRODCO to calculate the total costs and the RDT&E costs. Finally, the MAIN routine calls upon subroutine OUTPUT to print out the data in prescribed formats.

Except for an internal title designator, all output variables listed in COMMON are then set to zero, and a new set of input data is read in.* The above process is repeated over and over again until there are no more data sets to be read in, at which point the series of runs is terminated.

INPUT SPECIFICATIONS

As many sets of data as desired may be entered into the program, with each set constituting a run. The end of each set is signaled by a "flag" designator. After each set of data is read in, all of the computations, including printouts, are performed on that set before the machine reads in the next set of data.

* Before reading in the new set of data, if so designated by the CLEAR variable (relative address 027), the input data storage locations are also cleared. Otherwise, they are retained from that run.



* If no more sets of input data exist, run is terminated at this point.

Fig. 1--Flow of operations

Input Cards

The input data cards for the first of a series of runs must start with a separate title card, followed by the numerical-data cards for that run. The remaining sets may or may not start with a title card. This is an option reserved for the analyst. If a title card is not used, as indicated by a "flag" designator, the previous title is repeated for that run. It is important to note that the input data for each run must begin on a new card.

Each variable listed in COMMON in the program, regardless of whether it is an input or output variable, has an "address," or index number, in fixed-point (integer) form associated with it.* This "address" is the relative location of the variable in COMMON. That is, a variable with address 10 is located in the 10th cell in COMMON. Appendix C shows that for this program, the 10th cell location in COMMON is represented by the variable name RDAV, the avionics RDT&E cost. Since there are a total of 27 input variables listed first in COMMON in this program, their addresses are from 001 through 027. (Preceding zeros have no meaning in this case; they are used merely for convenience in keypunching so that all of the addresses can be expressed with three digits. The reason for this is that there are other computer cost programs at RAND whose input procedures are similar to the one described here. Hence, their input forms may be used for this program.) A listing of the input variables with their addresses is presented in Table 1.

In entering the data on cards, values for three input variables together with their corresponding addresses may be entered on each card. The address of each variable is entered in specified columns ("fields") immediately preceding those allotted to the variable. The "fields" reserved for the addresses and the "flag" designators are columns 1-5, 25-29, and 49-53. However, if three digits are used for the addresses, they must be entered specifically in columns 3-5, 27-29, and 51-53. The fields allotted to the corresponding variables are

* The old terms "fixed-point" and "floating-point" are now changed to "integer" and "real," respectively. However, in this Memorandum they are used interchangeably.

Table 1
INPUT VARIABLES

Address	Variable Name
001	Run number
002	Gross takeoff weight of aircraft (pounds)
003	Speed of aircraft (knots)
004	Maximum production rate (airframes per month)
005	Quantity at which maximum production rate is first achieved
006	Number of test aircraft
007	Engine-type designator (1, 2, 3, or 4) 1. Turbojet with afterburner 2. Turbojet with no afterburner 3. Turbofan 4. Turboprop
008	Thrust (pounds) per engine or equivalent shaft horsepower per engine
009	Number of engines per aircraft
010	Avionics RDT&E cost (millions of dollars)
011	Cost of avionics unit number 1 (millions of dollars)
012	Engine thrust designator (0, 1, 2, or 3) 0. All engine costs are to be calculated 1. Engine development cost (millions of dollars) is to be entered as a thrust 2. Cost (thousands of dollars) of first production-line engine is to be entered as a thrust 3. Both of the above engine costs are to be entered as thrusts
013	Engine development cost (millions of dollars) when entered as a thrust
014	Cost (thousands of dollars) of first production-line engine when entered as a thrust
015	Exponent of sine-power function in production-rate equation (if omitted, a value of .2. is assumed in the program)
016	Airframe-delete designator (if entered as 1., all airframe calculations and airframe printout are bypassed)
017	Engine-delete designator (if entered as 1., all engine calculations and engine printout are bypassed)
018	Avionics-delete designator (if entered as 1., all avionics calculations and avionics printout are bypassed)
019	Airframe profit factor (if omitted, a profit of 10 per cent (0.1) is assumed in the program)
020	Airframe engineering adjustment factor. <u>Note:</u> If 020 through 025 are omitted, each is assumed to have a value of 1. in the program
021	Airframe tooling adjustment factor

Table 1 -- continued

Address	Variable Name
022	Airframe labor adjustment factor
023	Airframe material adjustment factor
024	Engine development adjustment factor
025	Engine production adjustment factor
026	Turbofan weighting factor (if omitted, a value of 0.5 is assumed in the program so that turbofan cost is calculated as mean of turbojet costs)
027	Clear designator (if entered as 1., all input storage locations are set to zero before proceeding to next run)

columns 6-24, 30-48, and 54-72, respectively. Thus, 19 columns are reserved for each variable, with up to 9 allowed for decimals. Columns 73-80 are not processed by the machine and may be used as the analyst wishes, such as for sequencing. Figure 2 shows a data card with the specified fields for the variables and addresses.

Since all blank columns are read as zeros, care must be exercised in entering integer or fixed-point numbers (addresses and "flag" designators) onto the cards. The position of the right-hand digit of each such number must correspond with the right-hand column of each field specified for it. For example, consider columns 1-5, one of the fields specified for the addresses. Suppose that address 010 is keypunched in columns 2-4. Then, because column 5 remains (blank) in this field, the machine would read it as zero, and thus read the intended 10 as 100.

A floating-point (real) number, on the other hand, may be entered anywhere within the specified field since its keypunched decimal point overrides any decimal specification for that field. To enter a floating-point zero either a zero symbol is keypunched or else the field for the variable is left blank.

Figure 3 shows a typical card with data as used in this program. The variable corresponding with address 010 has a value of 5.0, the variable with address 004 has a value of 10., and the third variable with an address of 008 has a value of 18000.* As this card shows, the addresses do not necessarily have to be in order.** The reason for this is that the addresses are used as indexes. Thus, the variable corresponding to address 010 is stored in the tenth COMMON storage location, the value for address 004 is stored in the fourth COMMON storage location, etc.

Although the program provides for three sets of data to be keypunched on a card, as shown in Fig. 3, if any sets of fields are omitted, the machine will read those fields as zeros and continue to read sets

* Table 1 shows that the variables corresponding to these three addresses are: avionics RDT&E cost (010), maximum production rate (004), and thrust per engine (008).

** However, the variable and its address must always be kept together as a set.

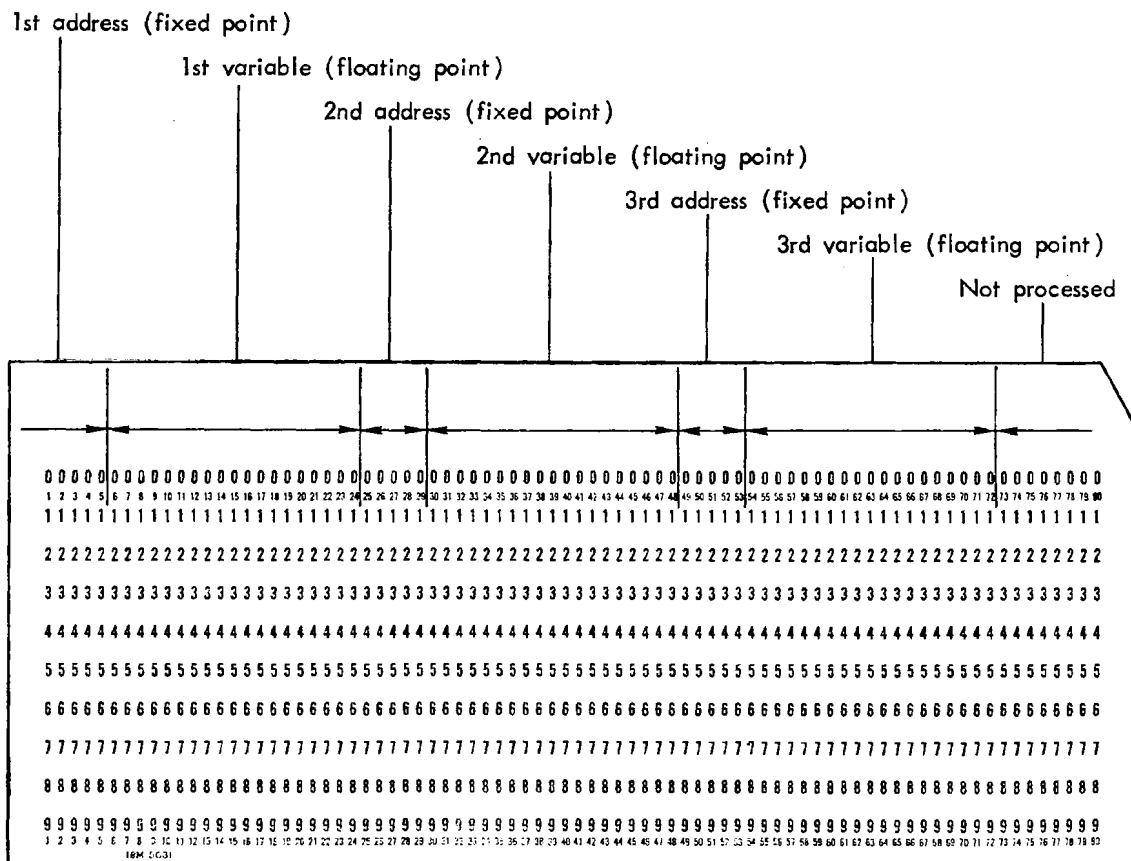


Fig. 2--Data card showing specified fields



Fig. 3--Data card showing sample inputs

of fields until it reads a "flag" designation.* However, in the ordering of the data, the blank sets are skipped since zero is an illegitimate index (subscript). As a result, if one wishes to change the value of, say, one variable on a card containing a set of three variables, he may do so by entering the new value with its address on a separate card placed anywhere behind the other card in that set of data. The new value will thus replace the previous value, since it will be read after the previous value is read.

If, for some reason, a variable is entered without an address (address of zero) or if an address less than zero or greater than 27 is entered, excluding the flag designators, an error message is printed and the series of runs is terminated at that point. The error message states that an address for an input variable has not been entered properly and that the job has been terminated.

Designators

The "flag" designator used to indicate the end of a set of data for each run is one of two types. One type is a set of four-digit, integer 5's (5555), entered in either columns 2-5, 26-29, or 50-53. This type indicates that the set of data for one run is completed and that a new title is to be read in and printed on a new page, before the next set of data is read in. Each title so entered must be keypunched on a separate card placed immediately before the set of data cards to which it refers. All 72 columns may be used for the title if necessary.

The other type of "flag" designator is a set of four-digit, integer 9's (9999), entered in any of the same columns as stated above. It also indicates that the set of data for one run is completed, but that the previous title is to be reprinted on a new page before reading in a new set of data. Thus, the "flag" designators serve two purposes. They signal the end of a set of data and also indicate whether a new title is to be read in and printed or whether the previous title is to be reprinted.

* If the machine goes through all of the cards and does not find a "flag" designation, control of the machine is returned to the FORTRAN monitor, thus ending the series of runs.

It should be noted again that the data for each run must always begin on a new card. Therefore, once a "flag" designator is entered on a card, the remaining portion of that card must be blank.

Either "flag" designator may be used to end the series of runs. In either case, the machine attempts to read in another set of data that does not exist, and, when this happens, control is returned to the FORTRAN monitor, ending the series of runs. When this happens, the statement "End-of-Data Encountered on System Input File" is printed, thus signifying the end of the series of runs.

PROGRAM STRUCTURE

A significant feature of this program is that the machine, unless specifically instructed to clear all locations, will automatically retain the input data for the next run. Unless such data are superseded by new data, they will be re-used. The significance of this feature is that after a complete set of data is entered for the first run, only changes in data need to be entered for succeeding runs. As a result, this program is suitably structured for sensitivity analyses, in which one may wish to study the effect of certain input variations on the output results.

An example of the above is depicted in Fig. 4, which shows one type of input sheet that may be used to enter the data for keypunching. Three sets of hypothetical data, each constituting a run, have been entered on this form. The first set (System A) ends with a "5555" because a new title is to be read in for the next run (System B). The first set for System B ends with a "9999" since the next run is to be a variation of the first run of System B and hence is to use the same title. For the variation run of System B (the last run of the series), the only variables that are changed are the run number (001), the air-frame material adjustment factor (023), and the engine development adjustment factor (024). Hence, these are the only variables that need to be entered on the form for that run. (The latter two sets are purposely entered out of order with respect to their addresses in order to show that data may be entered in this manner.)

[illegible]

Fig. 4--Example of input data form for DAPCA computer program

The input form shown in Fig. 4 has a fixed, decimal-point location for keypunching. However, for ease in keypunching, some users may prefer a form in which the values of the variables are entered left-justified so that they always start in the same column in each field. This is certainly permissible if the decimal point is always keypunched, since a keypunched decimal overrides any implied decimal point location. For this program, the implied decimal point locations are between columns 15 and 16, 39 and 40, and 63 and 64.

OPTIONS

Through the use of input designators, several options are provided for the analyst making a series of runs. These designators are shown in the input listing in Table 1. The variable corresponding to address 007 is used to designate the type of engine being considered. It must always be entered when the engine portion of the model is being used, since the engine cost equations depend on engine type.

The variable with address 012 is used to thruout a portion or all of the engine costs. For some runs, it may be desired to allow all of the engine costs to be calculated within the program. In this case, this variable is not used; that is, its value is left at zero. However, for other runs, it may be desired to input either the cost of the engine RDT&E or the production cost of the first production-line engine, or possibly both of these costs. Then, this variable is used as shown in the table.

The variable corresponding to address 015 is the exponent of the sine-power function used in calculating production rates for the sustaining tooling equation. Depending on the value of the exponent, this function is used to generate smooth production rate build-up curves of various shapes from zero to given maximum rates. It is of the following form:*

$$PR = PR_{\max} \cdot \sin^A \left(\frac{\pi}{2Q_{\max}} \cdot Q \right),$$

*The use of a trigonometric-type function for this program was conceived by Mr. R. L. Petruschell of the RAND Cost Analysis Department.

where PR = production rate (in airframes per month),*

$PR_{\max}$ = maximum production rate (in airframes per month),

A = exponent of sine function (address 015),

$\pi = 3.1415926$,

Q = airframe quantity,

$Q_{\max}$ = airframe quantity at which maximum production rate is first achieved.

The exponent A may be omitted in the input data. If it is, a value of 2 is assumed for A in the program, thus resulting in the sine-square curve shown in Fig. 5. The decision to use a sine-power function is completely arbitrary and is based only on the knowledge that, in general, an actual production rate buildup probably approximates some type of S-shaped curve.

For the computer program the production rate is rounded to the nearest tenth. However, if the production rate is calculated to be less than one airframe per month, it is reset to a value of one so that the minimum rate is one airframe per month. Also, if the airframe quantity becomes larger than $Q_{\max}$, it is reset to $Q_{\max}$ for use in the production rate equation. In this way, the curve stays up to $PR_{\max}$ once $PR_{\max}$ is reached.

The designators corresponding to addresses 016, 017, and 018 are used to bypass, respectively, the airframe, engine, or avionics portions of the model when they are not being used. For instance, if one is interested in calculating only airframe costs, by entering a 1. for both 017 and 018, the engine and airframe portions of the model including their printouts are bypassed. If any portion is bypassed, all inputs relating to that portion only do not have to be entered in the program.

Designator 019 is used to include an airframe profit factor. It may be omitted, in which case a profit of 10 percent (0.1) is assumed

* All letters taken together, such as PR , are to be treated as one variable as is the case in the computer program. However, for simplicity, the symbols used in the examples in the test may not always be the same as those used in the computer program.

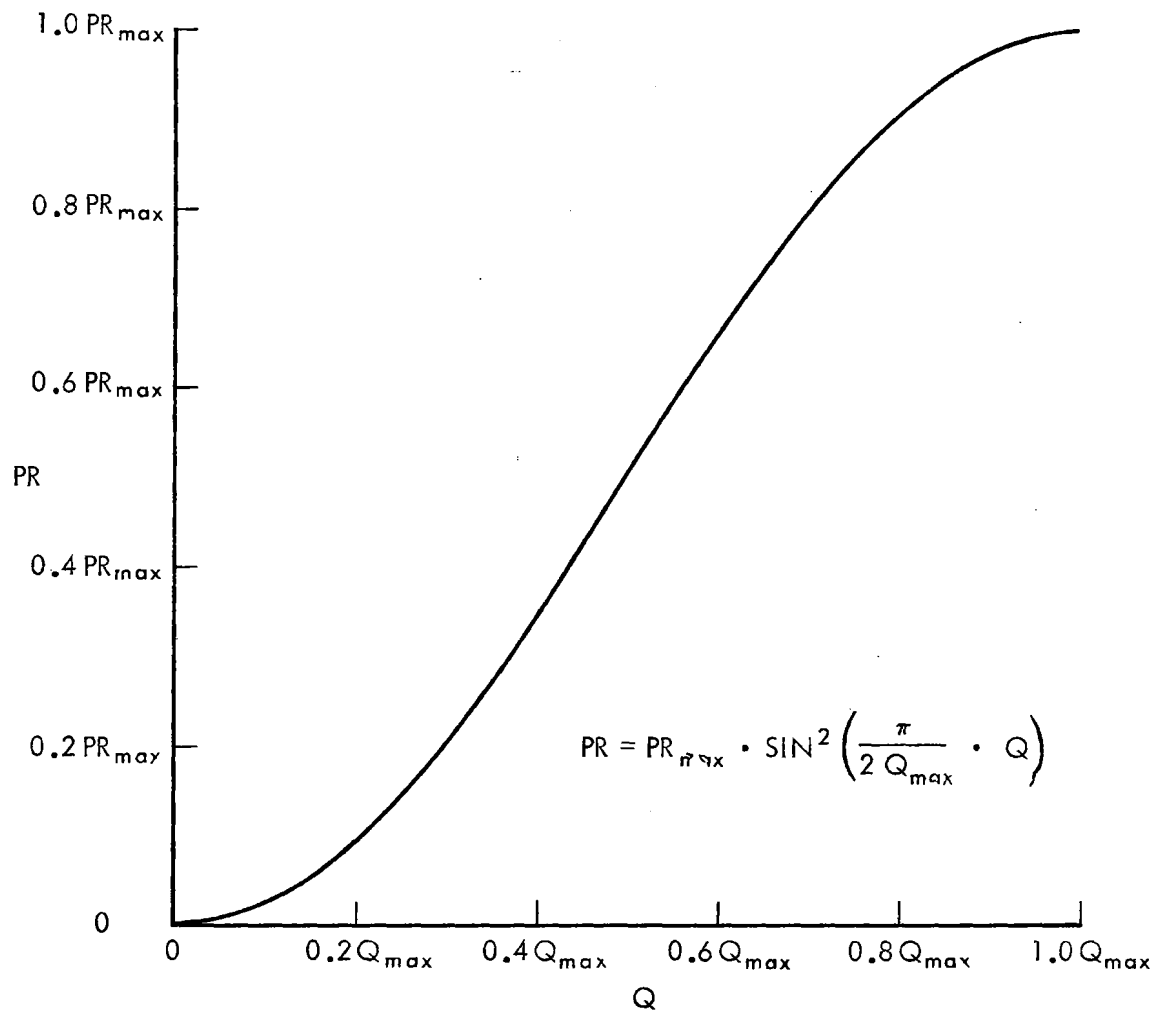


Fig. 5--Production rate curve based on sine-square function

for the program. The engine and avionics equations are assumed to include profits.

The designators corresponding to addresses 020 through 025 are adjustment factors for each of the categories as indicated. If the analyst wishes to adjust the production curves of any of the categories, he may do so by entering the adjustment as a decimal. For instance, if he feels the airframe tooling curve should be lowered 5 percent, he may do so by entering 0.95 for address 021. If any of these factors are omitted, a value of 1.0 is used for that factor in the program.

The turbofan engine production cost is calculated as a weighted average of the costs of turbojet engines with and without afterburners. By using the designator with address 026, the analyst can use any weighted average he chooses. This is done through the use of the following type of equation in the program.

$$C_{TF} = F \cdot C_{TJW} + (1.-F) \cdot C_{TJN} ,$$

where C_{TF} = production cost of turbofan engine,

C_{TJW} = production cost of turbojet engine with afterburner,

C_{TJN} = production cost of turbojet engine with no afterburner,

F = weighting factor (0. to 1.) corresponding to address 026.

If this variable is not used, a value of 0.5 is used for F in the program, thus giving equal weight to both turbojet costs.

Finally, the clear designator (027) is used to clear all storage locations in common except for the internal title designator (address 072) before proceeding to the next run. Whenever this designator is used, a complete set of data for the next run must be entered except for the possibility of a new title card. This designator does not affect the title designator.

III. PROGRAM CONSIDERATIONS

This section presents a general discussion of the two cost sub-routines--DEVCO and PRODCO--which utilize all of the cost equations listed in Appendix A. These subroutines are also shown in block diagram form in Appendix B.

SUBROUTINE DEVCO

After subroutine READ reads in and orders the data, subroutine DEVCO calculates all of the cost portions that are independent of quantity. As an example, the total airframe sustaining engineering cost is represented by the following equation (see Appendix A):

$$SEAC = 10^{0.90462} \cdot S^{0.54716} \cdot TT^{0.88} \cdot 12.25 \cdot DM \cdot CRI \cdot (QT^{0.2} - 1),$$

where SEAC = total airframe sustaining engineering cost (millions of dollars),*

S = maximum speed of aircraft at best altitude (knots),

TT = total maximum thrust per aircraft (cruise engines, sea-level static) (pounds),

DM = $(1 + \text{profit factor}) \cdot 10^{-6}$,

CRI = adjustment factor (0. to 1.),

QT = total aircraft quantity.

One may see that the product of the coefficient terms preceding the quantity within the parentheses is independent of the number of aircraft (QT) and is a constant for any given run. There are similar terms present in the equations for the other cost categories, and, in general, they are referred to as initial costs--actually for aircraft quantity one.

After the initial costs are calculated in subroutine DEVCO, they are applied to the production equations in subroutine PRODCO to generate

*The constand 12.25 is defined on p. 92.

the production costs for the prescribed aircraft quantities used in the program. This is done by calling subroutine PRODCO from subroutine DEVCO. After the production costs are calculated in subroutine PRODCO, the operation returns to subroutine DEVCO to determine the various RDT&E costs for airframes and, when so designated, for engines. As stated previously, the RDT&E cost for avionics is not calculated but is entered as a thruput in the program (address 010).

The total engine development cost, when not entered as a thruput, comprises the initial development cost, as calculated in subroutine DEVCO, plus a recurring engine development cost for the test aircraft and the first 50 production aircraft. This quantity of 50 production aircraft is an additional value. It is intended to reconcile those engine development expenditures that occur after the engine is qualified, but that are still associated with the aircraft.

The total airframe cost of test aircraft is also calculated in subroutine DEVCO. This cost includes the cost of the engines and avionics for the test aircraft as well as the cost of the airframes for the test aircraft.

SUBROUTINE PRODCO

As stated previously, the purpose of subroutine PRODCO is to determine the production costs of the major subsystems of the flyaway aircraft, i.e., airframes, engines, and avionics. The equations that are used to calculate these costs are listed in Appendix A. The aircraft production quantities used are 1 through 10 in steps of 1; 10 through 100 in steps of 10; 100 through 1,000 in steps of 100; 1,000 through 10,000 in steps of 1,000; and 20,000. These quantities exclude test aircraft, which are treated separately.

Basically, subroutine PRODCO operates as follows. Thirty-nine (39) passes are made through this subroutine. The first pass is made to determine the costs relating to the test aircraft. Therefore, for this first pass the quantity of aircraft used is the number of test aircraft being considered. (If there are no test aircraft included for the run, portions of this pass are bypassed, and the next pass is started.) The remaining 38 passes are made to calculate the various production costs

for the prescribed aircraft production quantities set forth above. For any given pass, the airframe costs are calculated first, the engine costs are calculated next, and the avionics costs are calculated last, before the totals are calculated. Upon completion of these calculations for a given pass, the quantity of aircraft is increased to its prescribed value, and the process is repeated.

For this program, the airframe costs consist of labor, material, sustaining tooling, and sustaining engineering costs. Except for airframe labor and material, the basic production cost equations for airframes, engines, and avionics represent cumulative-average costs for any quantity. The airframe labor and material equations represent unit costs and hence are treated differently. In general, the following procedure is used to calculate for each cost category, except airframe labor and material, the cumulative average and unit costs at any aircraft production quantity.

Airframe Costs

First, the total cost in each category is calculated from the basic equations for the number of test aircraft. For subsequent quantities of aircraft (passes 2-39), as set forth above, the total costs for these given quantities plus the test aircraft are calculated. The previously calculated costs for the test aircraft are then subtracted from these totals, the differences of which represent the total production costs. These costs are then divided by the total quantity minus the test aircraft, to give the cumulative-average costs at the production quantity being considered during this pass.

As an example, suppose there are five test aircraft whose total cost is \$25 million. That is,

$$C_{TA} = 25.$$

Next, assume that the total cost of the first seven aircraft including the test aircraft is \$31 million. That is,

$$C_{TA+2} = 31.$$

Then, the cumulative average production cost at the second production aircraft, excluding the test aircraft, is:

$$CA_2 = \frac{C_{TA+2} - C_{TA}}{(TA+2) - TA}$$

$$CA_2 = \frac{31 - 25}{2} = \$3 \text{ million.}$$

The unit production costs are calculated as follows. Once the total production costs are calculated for any quantity Q of aircraft including the test aircraft, the same equation is used to calculate the total costs for the same quantity of aircraft less one (Q - 1). The difference between these costs is the unit cost of the Qth aircraft. In calculating the unit airframe tooling cost, care must be taken in using the production-rate equation (sine-power function). Since this equation is a function of quantity, it must be recomputed for Q - 1 (if Q is not greater than the quantity at which the maximum production rate is first reached).

As mentioned previously, the airframe labor and material production costs are based on unit-cost equations; that is, the unit costs are calculated first. Factors are then applied to these costs to convert them to cumulative-average costs. For this computer program, these factors were derived as outlined below.

It may be seen from Appendix A that the equations used to calculate the unit material and labor costs are of the form:

$$C_u = A \cdot x^B,$$

where C_u = unit cost at quantity x,

A = unit cost at quantity 1,

x = quantity,

B = exponent.

Then, the cumulative average cost at any quantity Q is, by definition:

$$C_c = \frac{\sum_{x=1}^Q A \cdot x^B}{Q}, \quad (x = 1, 2, 3, \dots, Q) \quad (1)$$

Therefore, the ratio of C_c to C_u is:

$$\begin{aligned} \frac{C_c}{C_u} &= \alpha = \frac{\sum_{x=1}^Q A \cdot x^B}{Q \cdot A \cdot Q^B}, \\ \alpha &= \frac{\sum_{x=1}^Q x^B}{Q^{B+1}}. \end{aligned} \quad (2)$$

Although Equation (2) represents the expression for calculating the ratio of the cumulative average to unit cost at any quantity Q , it has the disadvantage of requiring the calculation of x^B for every point within the range being considered. Thus, for a quantity of 1000, the value of x^B for $x = 1, 2, 3, \dots, 1000$ would have to be calculated and summed. On the other hand, if the numerator in Equation (2) could be replaced by an integration expression, a final expression might be obtained that would eliminate all of the calculations between the starting and end points.* It can be shown that an integration expression may be used to approximate the numerator in Equation (2) if the beginning and end points for the integration are 0.5 and $Q + 0.5$, respectively.** Therefore,

* It must be remembered that for this problem x is restricted to discrete values (integers only).

** Harold Asher, Cost-Quantity Relationships in the Airframe Industry, The RAND Corporation, R-291, July 1956, pp. 34-38.

It may be shown that the beginning point for integration for exponential functions actually varies with Q . However, 0.5 gives sufficiently accurate answers, especially as Q becomes large.

$$\alpha = \frac{\int_{0.5}^{Q+0.5} x^B dx}{Q^{B+1}},$$

$$\alpha = \frac{x^{B+1} \Big|_{0.5}^{Q+0.5}}{(B+1) \cdot Q^{B+1}},$$

$$\alpha = \frac{(Q+0.5)^{B+1} - 0.5^{B+1}}{(B+1) \cdot Q^{B+1}}. \quad (3)$$

Equation (3) thus represents an approximate expression for calculating the ratio of cumulative average to unit costs for any given quantity Q and exponent B.

After the cumulative-average to unit cost ratios (α) are obtained during each pass for the aircraft quantity being considered, the total labor and material costs are obtained by multiplying the unit costs by their respective α 's and then by the aircraft quantity. Once these total costs are obtained, the cumulative-average production costs are obtained, as described previously, for the other cost items. That is, the previously calculated costs for the test aircraft are subtracted, respectively, from these totals to give the total production costs. These costs are then divided by the total aircraft quantity less test aircraft, to give the cumulative-average costs at the production quantity being considered during the pass.

Engine Costs

Since there may be more than one engine per aircraft, the engine costs are calculated as follows. First, it is assumed that, for multi-engine aircraft consisting of V number of engines per aircraft, the first set (V) of production engines is allotted to the first aircraft, the second set to the second aircraft, etc. If there are test aircraft, then with regard to production engines, the first test aircraft is

assumed to be fitted with the first set of production engines, etc. For example, assume that there are five 4-engine test aircraft. Then the first 5 x 4, or 20, production engines are allotted to these test aircraft, and the first production aircraft is assumed to be fitted with production engines 21 through 24. The second production aircraft is assumed to be fitted with engines 25 through 28, and so forth.

The total and cumulative-average engine costs for the aircraft are calculated as follows. After the total engine costs are calculated for the test aircraft, as described above, the total engine costs are calculated for the total aircraft quantity under consideration plus the test aircraft. These calculations are performed exactly as for the other items except that, if there are V number of engines per aircraft, the number of engines considered is V times the number of aircraft being considered.

In order that engine costs may be added to the airframe and avionics costs per aircraft, they must be calculated on a per aircraft basis. Therefore, the total engine production costs per aircraft are obtained as the sum of the unit costs of the engines on the aircraft. If there are V engines per aircraft, the sum of the engine unit costs per aircraft at aircraft quantity Q is obtained as follows:

$$E_{ACFT} = (E_x - E_{x-1}) + (E_{x-1} - E_{x-2}) + \dots (E_{x-V+1} - E_{x-V}),$$

$$E_{ACFT} = E_x - E_{x-V},$$

where E_{ACFT} = total engine production costs per aircraft at aircraft quantity Q,

E_x = total engine production costs for x engines,

E_{x-1} = total engine production costs for x-1 engines,

$\vdots$

E_{x-V} = total engine production costs for x-V engines,

V = number of engines per aircraft,

x = total number of production engines = V · Q.

Avionics

As stated previously, the avionics RDT&E cost is not calculated in the program, but is treated as a thruput. The total avionics recurring or production cost for aircraft quantity Q is of the form:

$$A_Q = A_1 \cdot Q^B,$$

where A_Q = total avionics production cost for aircraft quantity Q,

A_1 = avionics production cost at aircraft quantity 1 (input),

Q = total aircraft quantity,

B = exponent = .81558 for this program.

As the above equation suggests, it is assumed that the first avionics production package, or unit, is allotted to the first aircraft, the second to the second aircraft, etc. In the case of test aircraft, the first avionics package is allotted to the first test aircraft, etc. Therefore, as for airframes and engines, avionics costs allotted to test aircraft are subtracted from the totals to give the costs allotted to the production aircraft.

Total Costs

Total unit costs per aircraft are obtained simply by summing the airframe, engine, and avionics unit costs per aircraft for each aircraft quantity considered. For the cumulative-average costs, the total production costs for each cost component--namely, airframe labor, material, engineering, tooling; engines; and avionics--are summed and then divided by the number of production aircraft (total aircraft less test aircraft). This is done in this manner because these costs are already available from previous calculations.

This concludes the discussion on subroutine PRODCO.

IV. PROGRAM OUTPUTS

Figures 6 through 11 show the complete output for a typical run. The values are completely hypothetical and should not be interpreted as having any real meaning or representing any system either in existence or being proposed. The first page of the output (Fig. 6) provides an input listing for all of the inputs either entered in, or assumed by, the program.

The next page (Fig. 7) shows the airframe RDT&E costs. The top line shows a breakout of these costs, with the total being shown at the end of the second line. The third line is a breakout of the test aircraft cost shown on the top line. The total cost per airframe (at the end of the third line) times the quantity of test aircraft gives the test aircraft cost.

The next page (Fig. 8) lists the airframe unit and cumulative-average costs starting at aircraft production quantity 1 and continuing out to aircraft production quantity 20,000. All quantities and costs relating to the test aircraft have been subtracted out of the values shown in this figure. This is also true for all production costs relating to engines and avionics. Since there are 3 test aircraft for this run, quantity 20, for example, is actually the 23rd aircraft. Any unit cost at quantity 20 is actually the unit cost of the 23rd aircraft, whereas any cumulative-average cost at quantity 20 is the difference between the cost for 23 aircraft and the cost for the 3 test aircraft divided by the difference in quantity, $23 - 3$, or 20.

Figure 9 shows the engine RDT&E and production costs. With regard to the engine RDT&E cost, it might be noted again that a recurring development cost for the engines for the first 50 production aircraft is also included in this cost. In the example shown, there are 4 engines per aircraft, and the costs listed in the second and third columns are thus shown at 4 times the aircraft quantities. The engine unit and cumulative-average costs are also shown on an aircraft basis (fifth and sixth columns), and can thus be added directly to the airframe and avionics unit and cumulative-average costs, respectively, to give the total aircraft costs.

SYSTEM A
RUN 1

INPUT DATA

PRINCIPAL INPUTS

GROSS TAKEOFF WEIGHT OF AIRCRAFT (IN POUNDS)	200000.
SPEED OF AIRCRAFT (IN KNOTS)	500.
MAXIMUM PRODUCTION RATE (PRM)	10.
QUANTITY AT WHICH PRM IS FIRST ACHIEVED	50.
EXPONENT FOR PRODUCTION RATE EQUATION	2.00
NUMBER OF TEST AIRCRAFT	3.
NUMBER OF ENGINES PER AIRCRAFT	4.
THRUST PER ENGINE (IN POUNDS) -- TURBOFAN	18000.
WEIGHTING FACTOR FOR TURBOJET WITH AFTERBURNER	0.50
AVIONICS ROTE COST (IN MILLIONS OF DOLLARS)	5.000
AVIONICS COST FOR UNIT 1 (IN MILLIONS OF DOLLARS)	0.300
AIRFRAME PROFIT FACTOR	0.10

AIRFRAME ADJUSTMENT FACTORS

ENGINEERING	1.00
TOOLING	1.00
LABOR	1.00
MATERIAL	1.00

ENGINE ADJUSTMENT FACTORS

DEVELOPMENT	1.00
PRODUCTION	1.00

DESIGNATORS

ENGINE TYPE (ADDRESS 007)	3.
ENGINE THRUPUT (ADDRESS 012)	0.
AIRFRAME DELETE (ADDRESS 016)	0.
ENGINE DELETE (ADDRESS 017)	0.
AVIONICS DELETE (ADDRESS 018)	0.
CLEAR (ADDRESS 027)	0.

Fig. 6--First page of output (input data listing)

AIRFRAME RESEARCH AND DEVELOPMENT COSTS (INCLUDING PROFIT)
(MILLIONS OF DOLLARS)

INITIAL ENGINEERING	INITIAL TOOLING	DEVELOPMENT SUPPORT	TEST AIRCRAFT	FLIGHT TEST	ENGINES FOR TEST A/C
61.006	31.631	77.689	60.165	9.990	7.428

AVIONICS FOR
TEST A/C

0.735

TOTAL
R AND D

248.643

-27-

TEST AIRCRAFT PRODUCTION COSTS (INCLUDING PROFIT)
(MILLIONS OF DOLLARS)

NUMBER OF TEST A/C	SUSTAINING ENGINEERING	SUSTAINING TOOLING	LABOR	MATERIAL	TOTAL PER AIRFRAME
3	4.997	1.726	12.298	1.034	20.055

Fig. 7--Second page of output (airframe RDT&E costs)

AIRFRAME PRODUCTION COSTS (INCLUDING PROFIT)
(MILLIONS OF DOLLARS)

-- COST OF TEST AIRCRAFT EXCLUDED --

QUANTITY	UNIT COSTS.....				CUMULATIVE AVERAGE COSTS.....					
	SUSTAINING ENGINEERING	SUSTAINING TOOLING	LABOR	MATERIAL	TOTAL	SUSTAINING ENGINEERING	SUSTAINING TOOLING	LABOR	MATERIAL	TOTAL
1	4.501	1.491	8.584	0.899	15.475	4.501	1.491	8.584	0.899	15.475
2	3.674	1.198	7.825	0.866	13.562	4.087	1.344	8.215	0.883	14.529
3	3.126	1.006	7.255	0.840	12.227	3.767	1.232	7.896	0.869	13.764
4	2.733	0.871	6.805	0.818	11.228	3.508	1.141	7.624	0.856	13.130
5	2.437	0.769	6.438	0.800	10.445	3.294	1.067	7.388	0.845	12.594
6	2.204	0.691	6.131	0.785	9.810	3.112	1.004	7.178	0.835	12.130
7	2.016	0.627	5.869	0.771	9.283	2.956	0.950	6.992	0.826	11.724
8	1.861	0.561	5.641	0.759	10.547	2.819	1.117	6.823	0.817	11.577
9	1.730	0.524	5.441	0.748	13.160	2.698	1.576	6.669	0.810	11.753
10	1.618	0.395	5.263	0.738	11.014	2.590	1.758	6.529	0.802	11.679
20	1.011	0.298	4.154	0.670	8.815	1.911	2.569	5.554	0.750	10.784
30	0.753	0.234	3.576	0.631	7.309	1.559	2.577	4.973	0.716	9.825
40	0.608	0.975	3.204	0.603	5.390	1.336	2.350	4.570	0.690	8.946
50	0.513	0.361	2.937	0.583	4.394	1.179	2.012	4.266	0.671	8.128
60	0.446	0.310	2.734	0.566	4.057	1.062	1.732	4.025	0.654	7.474
70	0.396	0.273	2.572	0.552	3.794	0.970	1.526	3.828	0.641	6.964
80	0.357	0.244	2.439	0.540	3.581	0.895	1.367	3.661	0.629	6.553
90	0.326	0.221	2.326	0.530	3.404	0.834	1.241	3.518	0.618	6.212
100	0.300	0.203	2.230	0.521	3.254	0.782	1.138	3.394	0.609	5.922
200	0.174	0.113	1.682	0.465	2.434	0.503	0.643	2.652	0.549	4.347
300	0.126	0.080	1.425	0.435	2.066	0.384	0.460	2.281	0.515	3.641
400	0.101	0.062	1.266	0.414	1.843	0.316	0.363	2.046	0.493	3.217
500	0.084	0.051	1.154	0.399	1.689	0.271	0.301	1.878	0.475	2.926
600	0.073	0.044	1.071	0.387	1.575	0.239	0.259	1.750	0.462	2.710
700	0.064	0.039	1.005	0.377	1.485	0.215	0.228	1.648	0.450	2.541
800	0.058	0.034	0.951	0.369	1.412	0.196	0.204	1.564	0.441	2.404
900	0.053	0.031	0.906	0.362	1.351	0.180	0.185	1.493	0.432	2.290
1000	0.048	0.028	0.867	0.355	1.299	0.167	0.169	1.433	0.425	2.194
2000	0.028	0.016	0.651	0.316	1.011	0.102	0.095	1.087	0.379	1.663
3000	0.020	0.011	0.550	0.296	0.877	0.076	0.068	0.923	0.354	1.421
4000	0.016	0.009	0.488	0.282	0.794	0.061	0.053	0.822	0.338	1.274
5000	0.013	0.007	0.445	0.271	0.737	0.052	0.044	0.750	0.326	1.172
6000	0.012	0.006	0.413	0.263	0.693	0.045	0.038	0.697	0.316	1.096
7000	0.010	0.005	0.387	0.256	0.659	0.040	0.033	0.654	0.308	1.036
8000	0.009	0.005	0.366	0.251	0.631	0.037	0.030	0.619	0.301	0.987
9000	0.008	0.004	0.349	0.246	0.607	0.033	0.027	0.590	0.295	0.946
10000	0.008	0.004	0.334	0.242	0.587	0.031	0.025	0.565	0.290	0.911
20000	0.004	0.002	0.250	0.215	0.472	0.018	0.014	0.425	0.258	0.716

Fig. 8--Third page of output (airframe production costs)

ENGINE RESEARCH AND DEVELOPMENT COST (INCLUDING PROFIT)
(MILLIONS OF DOLLARS)

307.988

ENGINE PRODUCTION COSTS (INCLUDING PROFIT)
(MILLIONS OF DOLLARS)

-- COST OF ENGINES FOR TEST AIRCRAFT EXCLUDED --

ENGINE QUANTITY	UNIT COSTS (ENGINES)	CUM. AVG. COSTS (ENGINES)	AIRCRAFT QUANTITY	TOTAL UNIT COSTS (AIRCRAFT)	CUM. AVG. COSTS (AIRCRAFT)
4	0.520	0.528	1	2.110	2.110
8	0.505	0.519	2	2.042	2.076
12	0.493	0.512	3	1.989	2.047
16	0.483	0.505	4	1.945	2.021
20	0.474	0.500	5	1.909	1.999
24	0.467	0.495	6	1.878	1.979
28	0.460	0.490	7	1.851	1.961
32	0.455	0.486	8	1.827	1.944
36	0.449	0.482	9	1.805	1.929
40	0.445	0.479	10	1.786	1.914
80	0.412	0.453	20	1.653	1.810
120	0.393	0.436	30	1.576	1.743
160	0.380	0.423	40	1.521	1.693
200	0.370	0.414	50	1.480	1.654
240	0.361	0.405	60	1.447	1.622
280	0.354	0.399	70	1.419	1.595
320	0.349	0.393	80	1.395	1.571
360	0.343	0.388	90	1.374	1.550
400	0.339	0.383	100	1.356	1.532
800	0.310	0.353	200	1.240	1.411
1200	0.294	0.336	300	1.177	1.343
1600	0.283	0.324	400	1.134	1.295
2000	0.275	0.315	500	1.101	1.260
2400	0.269	0.308	600	1.076	1.231
2800	0.264	0.302	700	1.054	1.207
3200	0.259	0.297	800	1.036	1.187
3600	0.255	0.292	900	1.020	1.169
4000	0.252	0.288	1000	1.006	1.154
8000	0.230	0.264	2000	0.919	1.055
12000	0.218	0.250	3000	0.872	1.001
16000	0.210	0.241	4000	0.840	0.965
20000	0.204	0.234	5000	0.815	0.937
24000	0.199	0.229	6000	0.796	0.915
28000	0.195	0.224	7000	0.780	0.897
32000	0.192	0.220	8000	0.767	0.882
36000	0.189	0.217	9000	0.755	0.868
40000	0.187	0.214	10000	0.745	0.856
80000	0.171	0.196	20000	0.679	0.782

Fig. 9--Fourth page of output (engine RDT&E and production costs).

AVIONICS RESEARCH AND DEVELOPMENT COST (INCLUDING PROFIT)
(MILLIONS OF DOLLARS)

5.000

AVIONICS PRODUCTION COSTS (INCLUDING PROFIT
(MILLIONS OF DOLLARS)

--- COST OF AVIONICS FOR TEST AIRCRAFT EXCLUDED ---

QUANTITY	UNIT	CUMULATIVE AVERAGE
1	0.194	0.194
2	0.185	0.190
3	0.179	0.186
4	0.173	0.183
5	0.169	0.180
6	0.165	0.178
7	0.162	0.175
8	0.159	0.173
9	0.156	0.171
10	0.154	0.170
20	0.138	0.157
30	0.129	0.149
40	0.123	0.143
50	0.118	0.138
60	0.114	0.134
70	0.111	0.131
80	0.108	0.129
90	0.106	0.126
100	0.104	0.124
200	0.092	0.111
300	0.085	0.103
400	0.081	0.098
500	0.078	0.094
600	0.075	0.091
700	0.073	0.089
800	0.071	0.087
900	0.070	0.085
1000	0.068	0.083
2000	0.060	0.074
3000	0.056	0.068
4000	0.053	0.065
5000	0.051	0.062
6000	0.049	0.060
7000	0.048	0.059
8000	0.047	0.057
9000	0.046	0.056
10000	0.045	0.055
20000	0.039	0.048

Fig. 10--Fifth page of output (avionics RDT&E and production costs)

COMBINED COSTS (INCLUDING PROFIT) FOR AIRFRAMES, ENGINES, AND AVIONICS
(MILLIONS OF DOLLARS)

AIRCRAFT QUANTITY	PROD. RATE (AIRFRAMES PER MONTH)	ENGINE RATE	AVIONICS RATE	TOTAL RDTE COST	 PRODUCTION COSTS WITHOUT RATE			. PRODUCTION COSTS WITH RATE .	
					UNIT	CUM. AVG.	TOTAL	CUM. AVG.	GRAND TOTAL
1	1.0	17.779	17.779	561.631	17.779	17.779	17.779	579.411	579.411
2	1.0	15.790	16.795		15.790	16.795	33.590	595.222	595.222
3	1.0	14.394	15.997		14.394	15.997	47.990	609.621	609.621
4	1.0	13.346	15.335		13.346	15.335	61.339	622.771	622.771
5	1.0	12.523	14.773		12.523	14.773	73.865	635.436	635.436
6	1.0	11.853	14.287		11.853	14.287	85.720	647.351	647.351
7	1.0	11.296	13.860		11.296	13.860	97.017	658.648	658.648
8	1.1	12.533	13.694		12.533	13.694	109.551	671.182	671.182
9	1.4	15.122	13.853		15.122	13.853	124.674	686.305	686.305
10	1.6	12.953	13.763		12.953	13.763	137.628	699.259	699.259
20	4.4	10.607	12.751		10.607	12.751	255.013	816.644	816.644
30	7.4	9.013	11.716		9.013	11.716	351.489	913.120	913.120
40	9.5	7.034	10.782		7.034	10.782	431.291	992.923	992.923
50	10.0	5.992	9.921		5.992	9.921	496.039	1057.670	1057.670
60	10.0	5.618	9.230		5.618	9.230	553.818	1115.449	1115.449
70	10.0	5.323	8.690		5.323	8.690	608.322	1169.753	1169.753
80	10.0	5.084	8.253		5.084	8.253	660.201	1221.832	1221.832
90	10.0	4.884	7.888		4.884	7.888	709.914	1271.546	1271.546
100	10.0	4.714	7.578		4.714	7.578	757.798	1319.429	1319.429
200	10.0	3.767	5.868		3.767	5.868	1173.609	1735.240	1735.240
300	10.0	3.328	5.087		3.328	5.087	1526.022	2087.653	2087.653
400	10.0	3.058	4.611		3.058	4.611	1844.258	2405.989	2405.989
500	10.0	2.868	4.280		2.868	4.280	2139.980	2701.612	2701.612
600	10.0	2.726	4.032		2.726	4.032	2419.306	2980.238	2980.238
700	10.0	2.612	3.837		2.612	3.837	2685.938	3247.569	3247.569
800	10.0	2.519	3.678		2.519	3.678	2942.327	3503.358	3503.358
900	10.0	2.441	3.545		2.441	3.545	3190.202	3751.833	3751.833
1000	10.0	2.374	3.431		2.374	3.431	3430.838	3992.469	3992.469
2000	10.0	1.990	2.791		1.990	2.791	5582.867	6144.479	6144.479
3000	10.0	1.805	2.491		1.805	2.491	7471.641	8033.272	8033.272
4000	10.0	1.687	2.303		1.687	2.303	9213.595	9775.226	9775.226
5000	10.0	1.603	2.171		1.603	2.171	10856.586	11418.217	11418.217
6000	10.0	1.538	2.071		1.538	2.071	12426.241	12987.873	12987.873
7000	10.0	1.487	1.991		1.487	1.991	13938.331	14499.962	14499.962
8000	10.0	1.444	1.925		1.444	1.925	15403.435	15965.067	15965.067
9000	10.0	1.408	1.870		1.408	1.870	16829.097	17390.728	17390.728
10000	10.0	1.377	1.822		1.377	1.822	18220.935	18782.566	18782.566
20000	10.0	1.191	1.546		1.191	1.546	30924.960	31486.591	31486.591

Fig. 11--Sixth page of output (total aircraft RDT&E and production costs
and airframe production rate listing)

The avionics costs are shown in the next figure (Fig. 10). The avionics RDT&E thruput cost is shown at the top of the page and the production costs below. At the present time, studies are being conducted with regard to avionics costs, and as results are obtained, it is intended to incorporate them into the program.

The last output page (Fig. 11) lists the combined airframe, engine, and avionics costs. At the top of the page, the RDT&E costs are listed for each major category and for the total. The production costs are listed without and with the RDT&E costs. The last column, therefore, lists the grand total cost, including RDT&E, for the aircraft quantity listed. Figure 11 also includes a listing of the calculated airframe production rates (second column from left). As mentioned before, these rates are calculated from a sine-power function used in the program and are rounded to the nearest tenth.

Appendix A

BASIC EQUATIONS

This appendix presents the basic equations that are used in the FORTRAN-IV computer program. For the convenience of the reader, these equations are listed for each major category of cost (airframe components, engines, and avionics) and for the totals. All costs are calculated in millions of dollars. A listing of the constants and symbols is presented at the end of this appendix.

RDT&E COSTS

AIRFRAMES

Initial Engineering Cost

$$EI = 10^{0.90462} \cdot S^{0.54716} \cdot TT^{0.88} \cdot 12.25 \cdot DM \cdot CR1$$

Development Support Cost

$$DS = EI \cdot \frac{15.6}{12.25}$$

Initial Tooling Cost

$$TI = 10^{-0.91248} \cdot S^{1.07437} \cdot W^{0.83913} \cdot 10.55 \cdot DM \cdot CR2$$

Flight-Test Cost

$$FTC = 0.58 \cdot TA^{1.1} \cdot W^{0.80} \cdot S^{0.90} \cdot (DM)$$

Total Airframe Cost of Test Aircraft

$$CT = TA \cdot AFA_1 \quad (AFA_1 \text{ is calculated in next section on production costs.})$$

Total Airframe Cost of Test Aircraft Including Engines and Avionics
for Test Aircraft

$$CTE = CT + TCT + AVT$$

Total Airframe RDT&E Cost

$$RDA = EI + TI + DS + FTC + CTE$$

ENGINES

Initial Development Cost (If Not Entered as Input)

$$DEI_1 = \begin{cases} 0.13937 \cdot T^{0.74356} \cdot CR5, & (\text{turbojet or turbofan}) \\ 2.82917 \cdot ESH^{0.35497} \cdot CR5, & (\text{turboprop}) \end{cases}$$

Total Engine Development Cost

$$DEI = DEI_1 + EDAC$$

$$RDE = DEI$$

AVIONICS

$$RDAV = \text{Input}$$

TOTALS

Total RDT&E Cost (Airframes, Engines, and Avionics)

$$RDT = RDA + RDE + RDAV$$

Grand Total Cost of Aircraft for Airframe Quantity QT (Airframes,
Engines, and Avionics)

$$TCOST_I = TC_I + RDT, \quad (I > 1)$$

PRODUCTION COSTS

AIRFRAMES

Unit Labor for Airframe Number 1

$$PAL = 6.762 \cdot 10^{0.16314} \cdot W^{0.73672} \cdot S^{0.43113} \cdot 9.50 \cdot 1.11 \\ \cdot 1.14 \cdot DM \cdot CR3$$

Unit Material for Airframe Number 1

$$PAM = 2.169 \cdot 10^{-0.76558} \cdot W^{0.77933} \cdot S^{0.856650} \cdot DM \cdot CR4$$

Cumulative Average to Unit Ratio for Airframe Labor

$$BLA_I = \frac{(QT + 0.5)^{0.585} - 0.5^{0.585}}{QT^{0.585} \cdot 0.585}$$

Cumulative Average to Unit Ratio for Airframe Material

$$BMA_I = \frac{(QT + 0.5)^{0.832} - 0.5^{0.832}}{QT^{0.832} \cdot 0.832}$$

Unit Labor Cost at Quantity QT (Total Aircraft Including Test Aircraft)

$$ULA_I = PAL \cdot QT^{-0.415}$$

Total Labor Cost for Aircraft Quantity QT

$$CALAC = ULA_I \cdot BLA_I \cdot QT$$

Cumulative Average Labor Cost for Test Aircraft

$$CALA_I = \frac{CALAC_{TA}}{QT}, \quad (I = 1, QT = TA)$$

Cumulative Average Labor Cost for Aircraft Quantity QT Less Test Aircraft

$$CALA_I = \frac{CALAC_{QT} - CALAC_{TA}}{QT - TA}, \quad (I > 1, QT > TA)$$

Unit Material Cost at Aircraft Quantity QT

$$UMA_I = PAM \cdot QT^{-0.168}$$

Total Material Cost for Aircraft Quantity QT

$$CAMAC = UMA_I \cdot BMA_I \cdot QT$$

Cumulative Average Material Cost for Test Aircraft

$$CAMA_I = \frac{CAMAC_{TA}}{QT}, \quad (I = 1, QT = TA)$$

Cumulative Average Material Cost for Aircraft Quantity QT Less Test Aircraft

$$CAMA_I = \frac{CAMAC_{QT} - CAMAC_{TA}}{QT - TA}, \quad (I > 1, QT > TA)$$

Airframe Production Rate Calculation

$$Q2 = QT$$

If $QT > QM$, $Q2$ is set equal to QM

$$QTEMP = \frac{\pi}{2 \cdot QM} \cdot Q2$$

$$PR_I = PRM \cdot \sin^2(QTEMP), \quad (PR_I \text{ is rounded to nearest tenth})$$

If $PR_I < 1$, PR_I is set equal to 1

Total Sustaining Tooling Cost for Aircraft Quantity QT

$$STAC = (PR_I^{0.4} \cdot QT^{0.138} - 1) \cdot TI$$

Cumulative Average Sustaining Tooling Cost for Test Aircraft

$$STA_I = \frac{STAC_{TA}}{QT}, \quad (I = 1, QT = TA)$$

Cumulative Average Sustaining Tooling Cost for Aircraft Quantity QT
Less Test Aircraft

$$STA_I = \frac{STAC_{QT} - STAC_{TA}}{QT - TA}, \quad (I > 1, QT > TA)$$

Total Sustaining Tooling Cost for Aircraft Quantity QT - 1 (Q1)

$$Q1 = QT - 1$$

$$STA1 = (PR_I^{0.4} \cdot Q1^{0.138} - 1) \cdot TI, \quad (PR_I \text{ is recalculated for } Q1)$$

Unit Sustaining Tooling Cost at Aircraft Quantity QT

$$STU_I = STAC - STA1$$

Total Sustaining Engineering Cost for Aircraft Quantity QT

$$SEAC = EI \cdot (QT^{0.2} - 1)$$

Cumulative Average Sustaining Engineering Cost for Test Aircraft

$$SEA_I = \frac{SEAC_{TA}}{QT}, \quad (I = 1, QT = TA)$$

Cumulative Average Sustaining Engineering Cost for Aircraft Quantity
QT Less Test Aircraft

$$SEA_I = \frac{SEAC_{QT} - SEAC_{TA}}{QT - TA}, \quad (I > 1; QT > TA)$$

Total Sustaining Engineering Cost for Aircraft Quantity QT - 1

$$SEA_1 = EI \cdot (Q_1^{0.2} - 1)$$

Unit Sustaining Engineering Cost at Aircraft Quantity QT

$$SEU_I = SEAC - SEA_1$$

Total Cumulative Average Cost for Test Aircraft (I = 1) or for Aircraft Quantity QT Less Test Aircraft (I > 1)

$$AFA_I = SEA_I + STA_I + CALA_I + CAMA_I$$

Total Unit Cost at Aircraft Quantity QT

$$AFU_I = SEU_I + STU_I + ULA_I + UMA_I$$

ENGINES

Quantity of Engines Required for Aircraft Quantity QT

$$X_I = V \cdot QT$$

Total Engine Recurring Development Cost for Aircraft Quantity QT

$$EDAC = \begin{cases} DEI \cdot (X_I^{0.07751} - 1), & (\text{turbojet or turbofan}) \\ DEI \cdot (X_I^{0.09334} - 1), & (\text{turboprop}) \end{cases}$$

Production Cost of Engine Number 1 (If Not Entered as Input)

$$CAPE = \begin{cases} 0.18700 \cdot T^{0.84845} \cdot CR6, & (\text{turbojet with afterburner}) \\ 0.31979 \cdot T^{0.81626} \cdot CR6, & (\text{turbojet with no afterburner}) \\ (TFW \cdot 0.18700 \cdot T^{0.84845} + TFN \cdot 0.31979 \cdot T^{0.81626}) \cdot CR6, & (\text{turbofan}) \\ 4.86224 \cdot ESH^{0.45873} \cdot CR6, & (\text{turboprop}) \end{cases}$$

Total Engine Production Cost for Aircraft Quantity QT

$$PAC = \begin{cases} CAPE \cdot X_I^{0.86745}, & (\text{turbojet with afterburner}) \\ CAPE \cdot X_I^{0.87088}, & (\text{turbojet with no afterburner}) \\ CAPE \cdot (TFW \cdot X_I^{0.86745} + TFN \cdot X_I^{0.87088}), & (\text{turbofan}) \\ CAPE \cdot X_I^{0.89055}, & (\text{turboprop}) \end{cases}$$

Engine Cumulative Average Production Cost for Test Aircraft

$$PA_I = \frac{PAC_{TA}}{X_I}, \quad (I = 1, X_I = V \cdot TA)$$

Engine Cumulative Average Production Cost for Aircraft Quantity QT

Less Test Aircraft

$$PA_I = \frac{PAC_{QT} - PAC_{TA}}{V \cdot (QT - TA)}, \quad (I > 1, QT > TA)$$

Total of Engine Recurring Development and Production Cost for Test Aircraft

$$TCT = EDAC$$

Total Engine Production Cost for Engine Quantity $X_I - 1$ (X_1)

$$X_1 = X_I - 1$$

$$PA_1 = \begin{cases} CAPE \cdot X_1^{0.86745}, & (\text{turbojet with afterburner}) \\ CAPE \cdot X_1^{0.87088}, & (\text{turbojet with no afterburner}) \\ CAPE \cdot (TFW \cdot X_1^{0.86745} + TFN \cdot X_1^{0.87088}), & (\text{turbofan}) \\ CAPE \cdot X_1^{0.89055}, & (\text{turboprop}) \end{cases}$$

Engine Unit Production Cost at Engine Quantity X_I

$$PU_I = PAC - PA_1$$

Total Engine Unit Production Cost per Aircraft at Aircraft Quantity QT

$$\begin{aligned} UEA_I &= (PAC - PA_{X_I-1}^1) + (PA_{X_I-1}^1 - PA_{X_I-2}^1) \\ &\quad + (PA_{X_I-2}^1 - PA_{X_I-3}^1) + \dots (PA_{X_I-(V-1)}^1 - PA_{X_I-V}^1) \\ UEA_I &= PAC - PA_{X_I-V}^1 \end{aligned}$$

AVIONICS

Total Avionics Production Cost for Aircraft Quantity QT

$$AVAC = AVI \cdot QT^{0.81558}$$

Avionics Cumulative Average Production Cost for Test Aircraft

$$AVC_I = \frac{AVAC_{TA}}{QT}, \quad (I = 1, QT = TA)$$

Avionics Cumulative Average Production Cost for Aircraft Quantity QT
Less Test Aircraft

$$AVC_I = \frac{AVAC_{QT} - AVAC_{TA}}{QT - TA}, \quad (I > 1, QT > TA)$$

Total Avionics Production Cost for Aircraft Quantity QT - 1 (Q1)

$$AVA1 = AVI \cdot Q1^{0.81558}, \quad (QT > 1)$$

Avionics Unit Production Cost at Aircraft Quantity QT

$$AVU_I = AVAC - AVA1$$

TOTALS

Total Unit Cost at Aircraft Quantity QT (Airframes, Engines, and Avionics)

$$TUN_I = AFU_I + UEA_I + AVU_I$$

Total Cumulative Average Cost for Test Aircraft (Airframes, Engines, and Avionics)

$$TCA_I = \frac{(CALAC + CAMAC + STAC + SEAC + PAC + AVAC)_{TA}}{QT}, \quad (I = 1, QT = TA)$$

Total Cumulative Average Production Cost for Aircraft Quantity QT
Less Test Aircraft (Airframes, Engines, and Avionics)

$$C_{QT} = (CALAC + CAMAC + STAC + SEAC + PAC + AVAC)_{QT}$$

$$C_{TA} = (CALAC + CAMAC + STAC + SEAC + PAC + AVAC)_{TA}$$

$$TCA_I = \frac{C_{QT} - C_{TA}}{QT - TA}, \quad (I > 1, QT > TA)$$

Total Aircraft Cumulative Average Cost Including RDT&E for Quantity QT
(Airframes, Engines, and Avionics)

$$TCQ = \frac{TCOST_I}{QT}$$

Total Production Cost for Test Aircraft

$$TC_I = TCA_I \cdot QT, \quad (I = 1, QT = TA)$$

Total Production Cost for Aircraft Quantity QT Less Test Aircraft

$$TC_I = TCA_I \cdot (QT - TA), \quad (I > 1, QT > TA)$$

CONSTANTS

12.25 = Engineering direct labor -- overhead, G&A, and miscellaneous direct charges in 1965 constant dollars.

15.6 = Dollars per engineering hour required for development support.

10.55 = Tooling direct labor -- overhead, G&A, miscellaneous direct charges, and material in 1965 constant dollars.

9.50 = Production direct labor -- overhead, G&A, and miscellaneous direct charges.

1.11 = Engineering changes as a percentage of production costs.

1.14 = Quality control as a percentage of production costs.

6.762 = Conversion factor from unit 100 to unit 1.

2.169 = Conversion factor from unit 100 to unit 1.

LIST OF SYMBOLS USED IN EQUATIONS

INPUTS

<u>Symbol</u>	<u>Meaning</u>
AVI	Avionics production cost of unit number one.
CR1	Adjustment factor for airframe initial engineering cost.
CR2	Adjustment factor for airframe initial tooling cost.
CR3	Adjustment factor for airframe labor cost.
CR4	Adjustment factor for airframe material cost.
CR5	Adjustment factor for engine initial development cost.
CR6	Adjustment factor for production cost of engine number one.
DM	$10^{-6} \cdot (1 + P)$. For P, see below.
ESH	Equivalent shaft horsepower per engine.
P	Airframe profit factor.
PRM	Maximum airframe production rate (airframes per month).
QM	Quantity at which maximum airframe production rate is first achieved.
RDAV	Avionics RDT&E cost (millions of dollars).
S	Maximum speed of aircraft at best altitude (knots).
T	Maximum thrust (pounds) per engine (cruise engine, sea-level static).
TA	Number of test aircraft.
TFW	Turbofan weighting factor, applied to cost of turbojet with afterburner.
V	Number of engines per aircraft.
W	Gross take-off weight of aircraft (pounds).

OUTPUTS

<u>Symbol</u>	<u>Meaning</u>
AFA _I	Total airframe cumulative average cost.
AFU _I	Total unit cost at aircraft quantity QT.
AVAC	Total avionics production cost for aircraft quantity QT.
AVC _I	Avionics cumulative average production cost.
AVA1	Total avionics production cost for aircraft quantity QT - 1.
AVT	Total avionics production cost for test aircraft.
AVU _I	Avionics unit production cost at aircraft quantity QT.
BLA _I	Ratio of cumulative average cost to unit cost for airframe labor.
BMA _I	Ratio of cumulative average cost to unit cost for airframe material.
CALA _I	Airframe cumulative average labor cost.
CALAC	Total airframe labor cost for quantity QT.
CAMA _I	Airframe cumulative average material cost.
CAMAC	Total airframe material cost for quantity QT.
CAPE	Production cost of engine number one.
CT	Total airframe cost of test aircraft.
CTE	Total airframe cost of test aircraft including engines and avionics for test aircraft.
DEI ₁	Engine initial development cost (may also be entered as an input).
DEI	Total engine development cost.
DS	Airframe development support cost.
EDAC	Total engine recurring development cost.
EI	Airframe initial engineering cost.

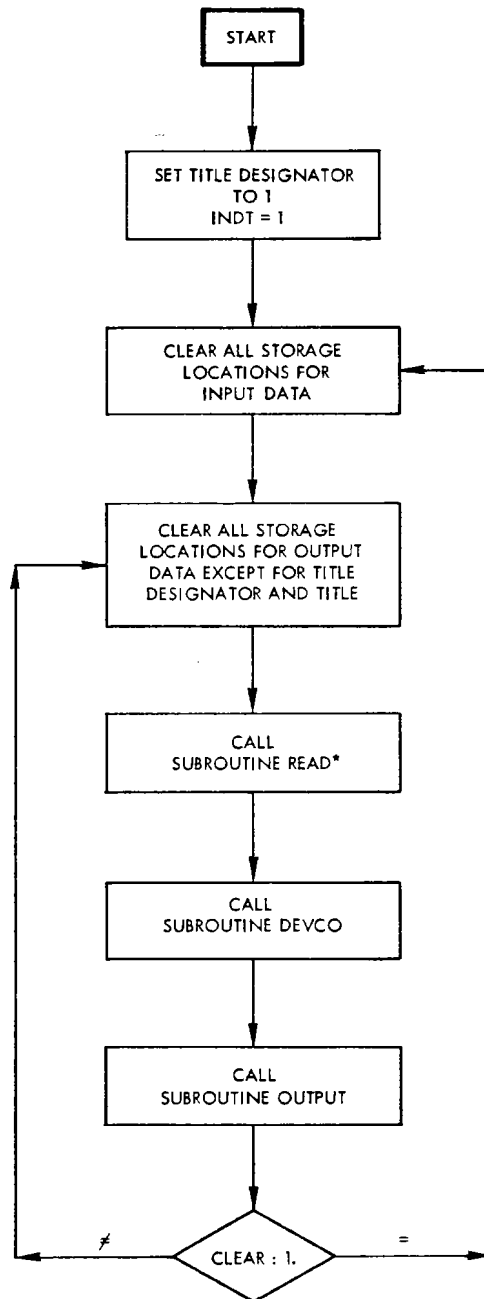
<u>Symbol</u>	<u>Meaning</u>
FTC	Airframe flight-test cost.
I	Used as subscript to some of the variables. (For test aircraft, $I = 1$; for any aircraft quantity QT less test aircraft, $I > 1$.)
PA_I	Engine cumulative average production cost.
PAC	Total engine production cost for aircraft quantity QT.
PA1	Total engine production cost for aircraft quantity QT - 1.
PAL	Unit labor cost for airframe number one.
PAM	Unit material cost for airframe number one.
PR_I	Airframe production rate (airframes per month).
PU_I	Engine unit production cost at engine quantity X_I .
QT	Total aircraft quantity (for $I = 1$, $QT = TA$; for $I > 1$, $QT > TA$).
Q1	QT - 1.
Q2	Set equal to QT. However, if $QT > QM$, Q2 is set equal to QM.
QTEMP	$\frac{\pi}{2 \cdot QM} \cdot Q2$.
RDA	Total airframe RDT&E cost.
RDE	Total engine development cost (set equal to DEI).
RDT	Total aircraft RDT&E cost (airframes, engines, and avionics).
SEA_I	Airframe cumulative average sustaining engineering cost.
SEAC	Total airframe sustaining engineering cost for quantity QT.
SEA1	Total airframe sustaining engineering cost for quantity QT - 1.
SEU_I	Airframe unit sustaining engineering cost at quantity QT.

<u>Symbol</u>	<u>Meaning</u>
SIN	Sine function.
STA _I	Airframe cumulative average sustaining tooling cost.
STAC	Total airframe sustaining tooling cost for quantity QT.
STAl	Total airframe sustaining tooling cost for quantity QT - 1.
STU _I	Airframe unit sustaining tooling cost at quantity QT.
TC _I	Total aircraft production cost (airframes, engines, and avionics).
TCA _I	Total aircraft cumulative average cost (airframes, engines, and avionics).
TCOST _I	Grand total cost of aircraft for quantity QT (RDT&E and production costs for airframes, engines, and avionics).
TCQ	Total aircraft cumulative average cost including RDT&E for quantity QT (airframes, engines, and avionics).
TCT	Total of engine recurring development cost and production cost for test aircraft.
TFN	Turbofan weighting factor, applied to cost of turbojet with no afterburner (TFN = 1.0 - TFW).
TI	Airframe initial tooling cost.
TT	Total maximum thrust (pounds) per aircraft (cruise engines, sea-level static) (V · T).
TUN _I	Total aircraft unit cost at quantity QT (airframes, engines, and avionics).
ULA _I	Airframe unit labor cost at quantity QT.
UMA _I	Airframe unit material cost at quantity QT.
X _I	Quantity of engines required for aircraft quantity QT (V · QT).
ZPT	Total engine unit production cost per aircraft at aircraft quantity QT.

Appendix B

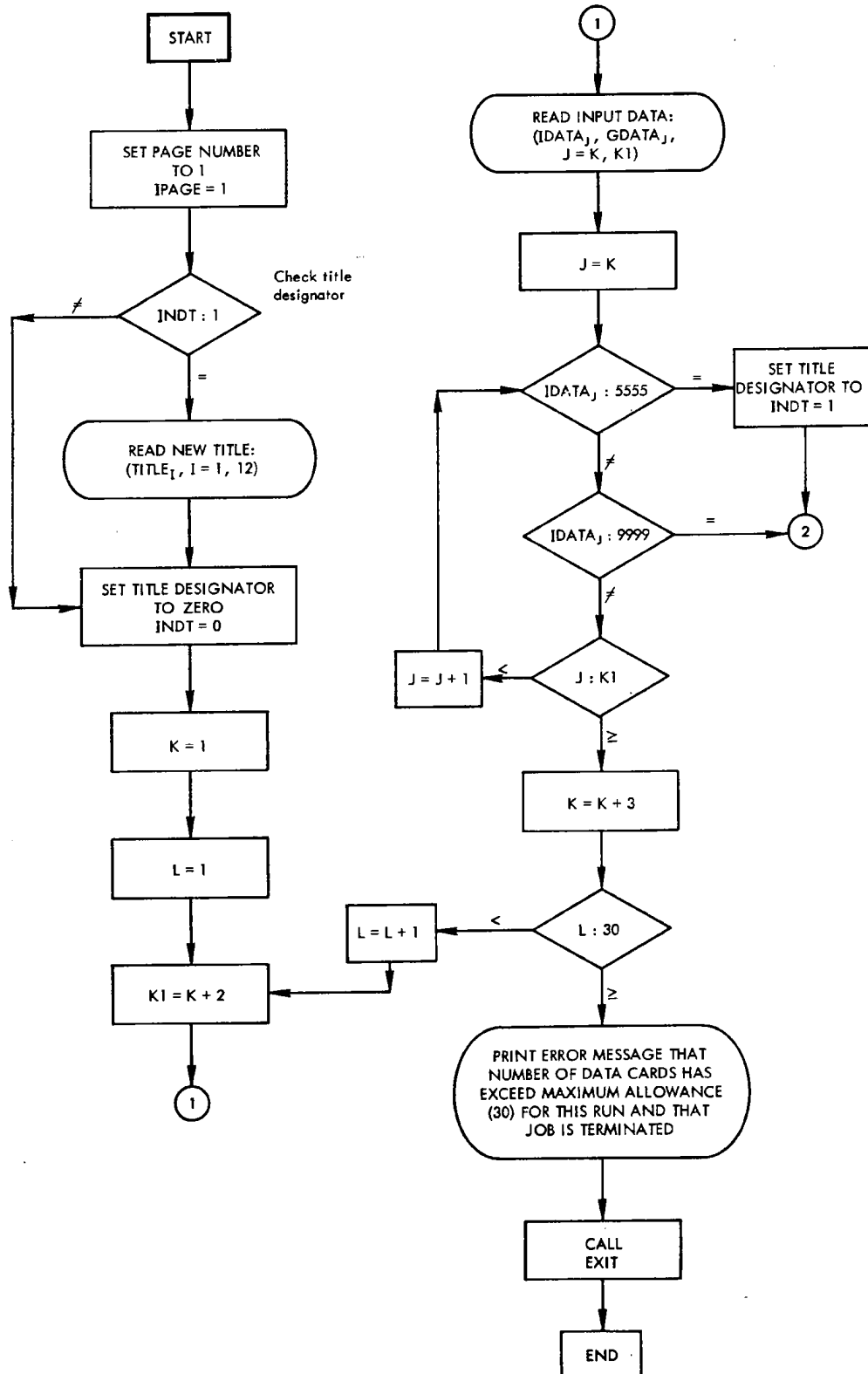
FLOW CHARTS OF PROGRAM

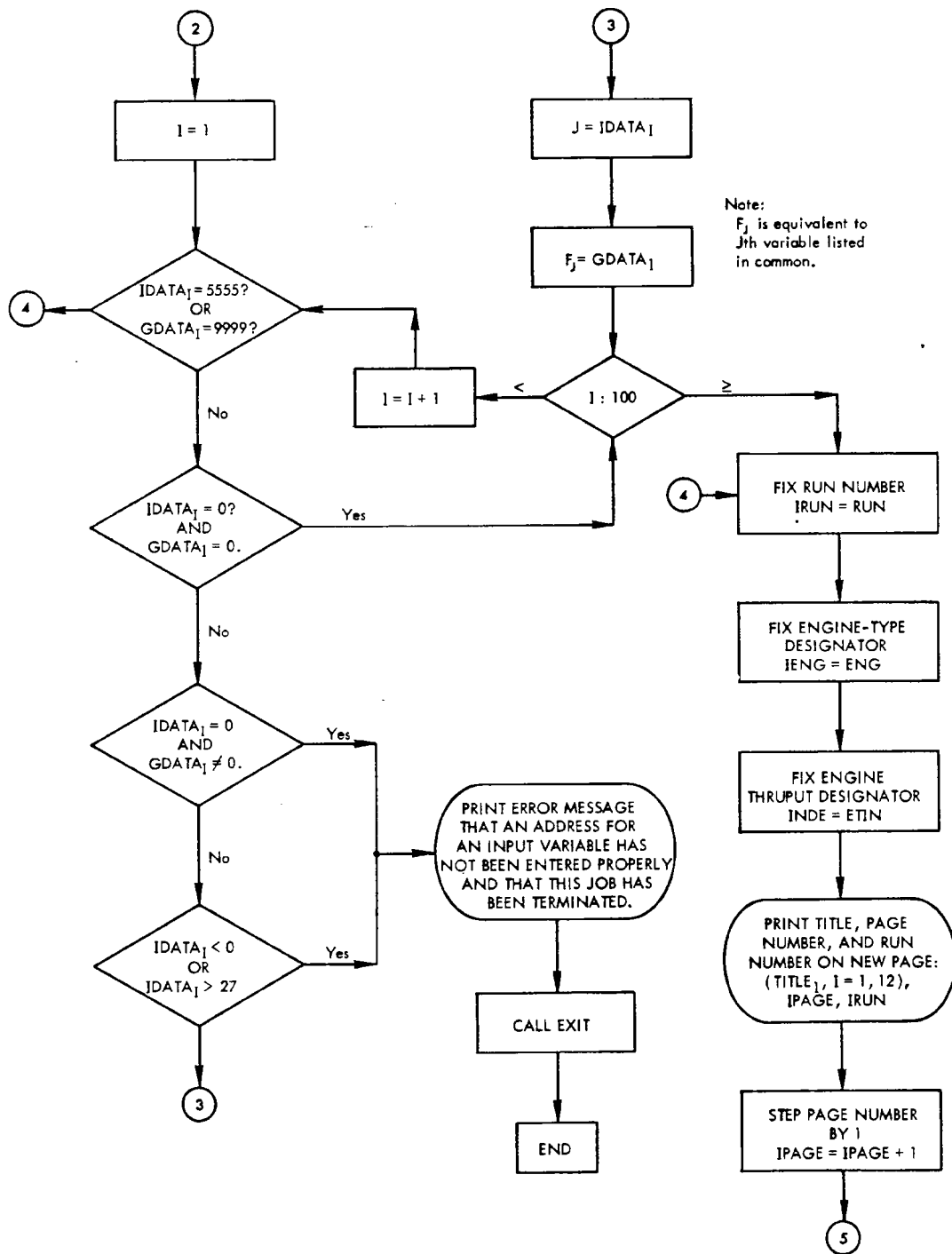
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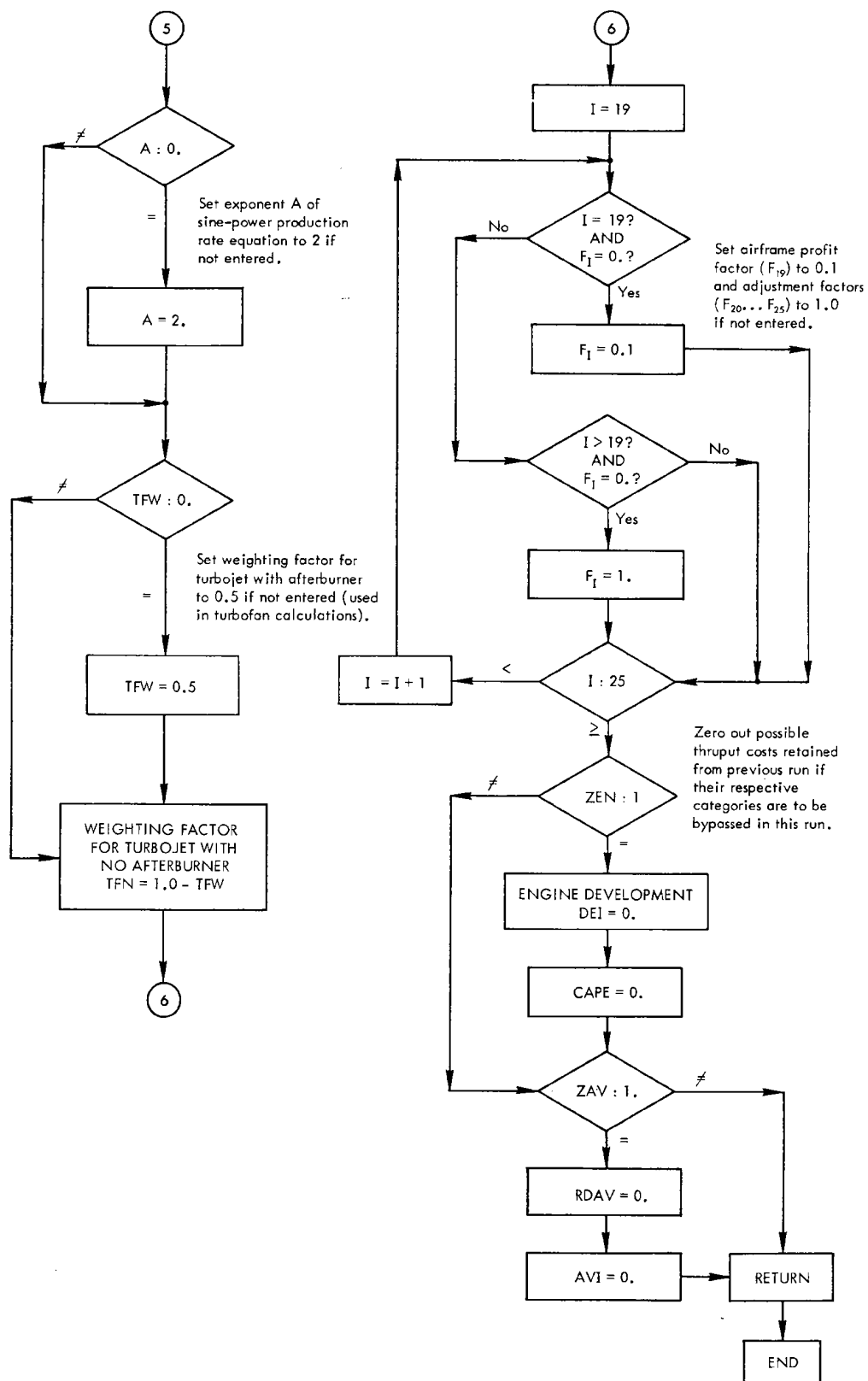


If no more sets of input data exist, run is terminated at this point.

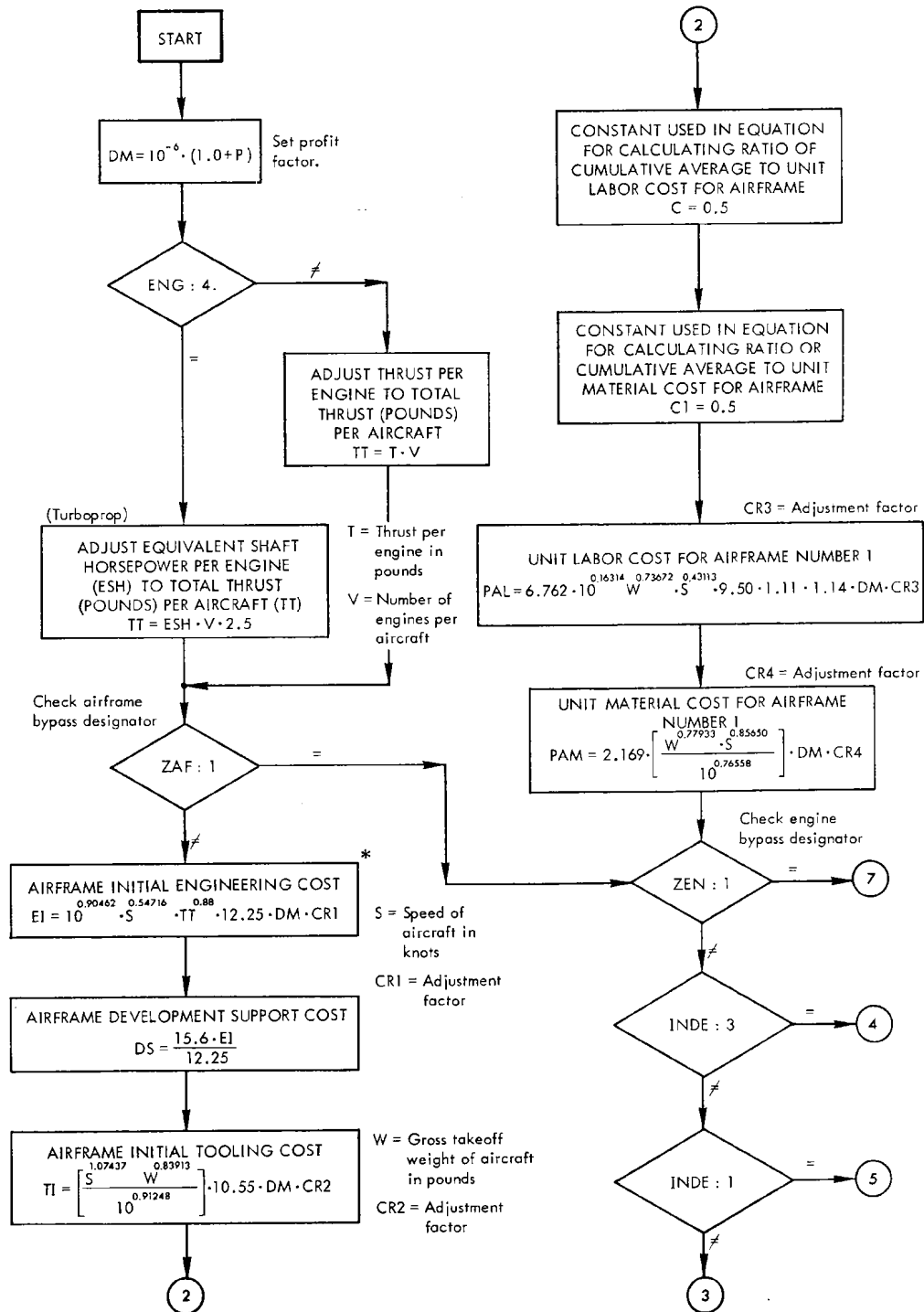
Subroutine READ



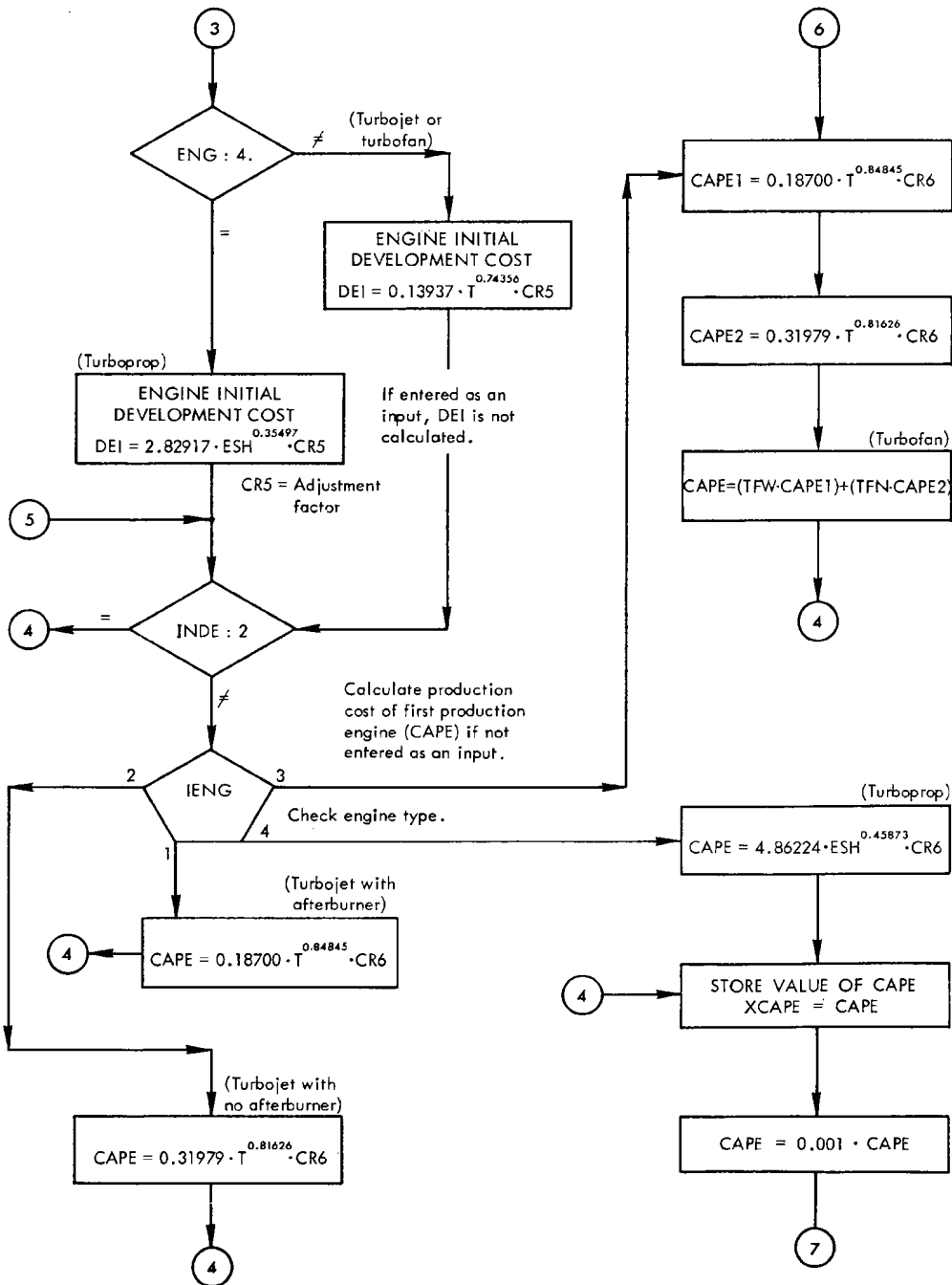


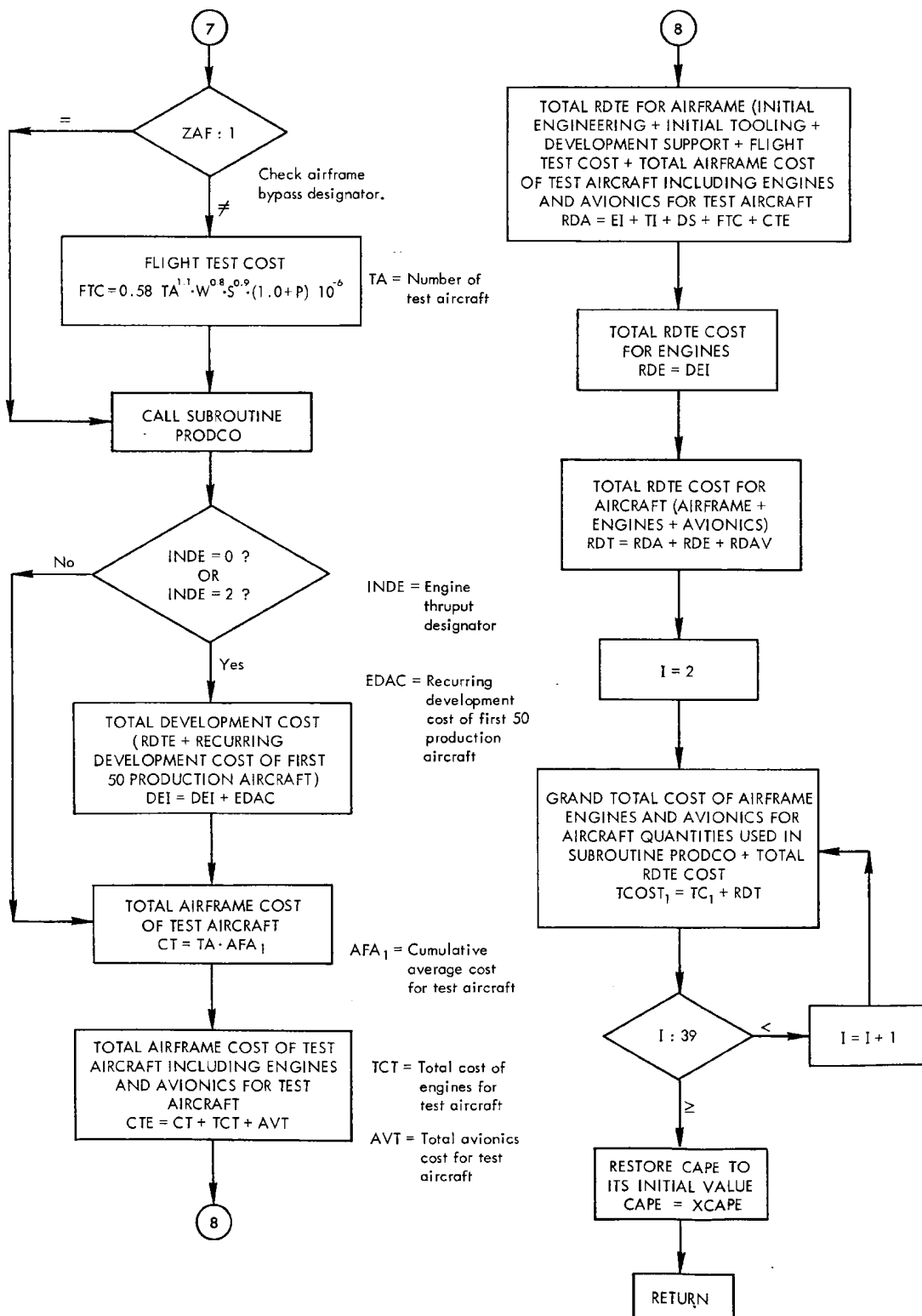


Subroutine DEVCO

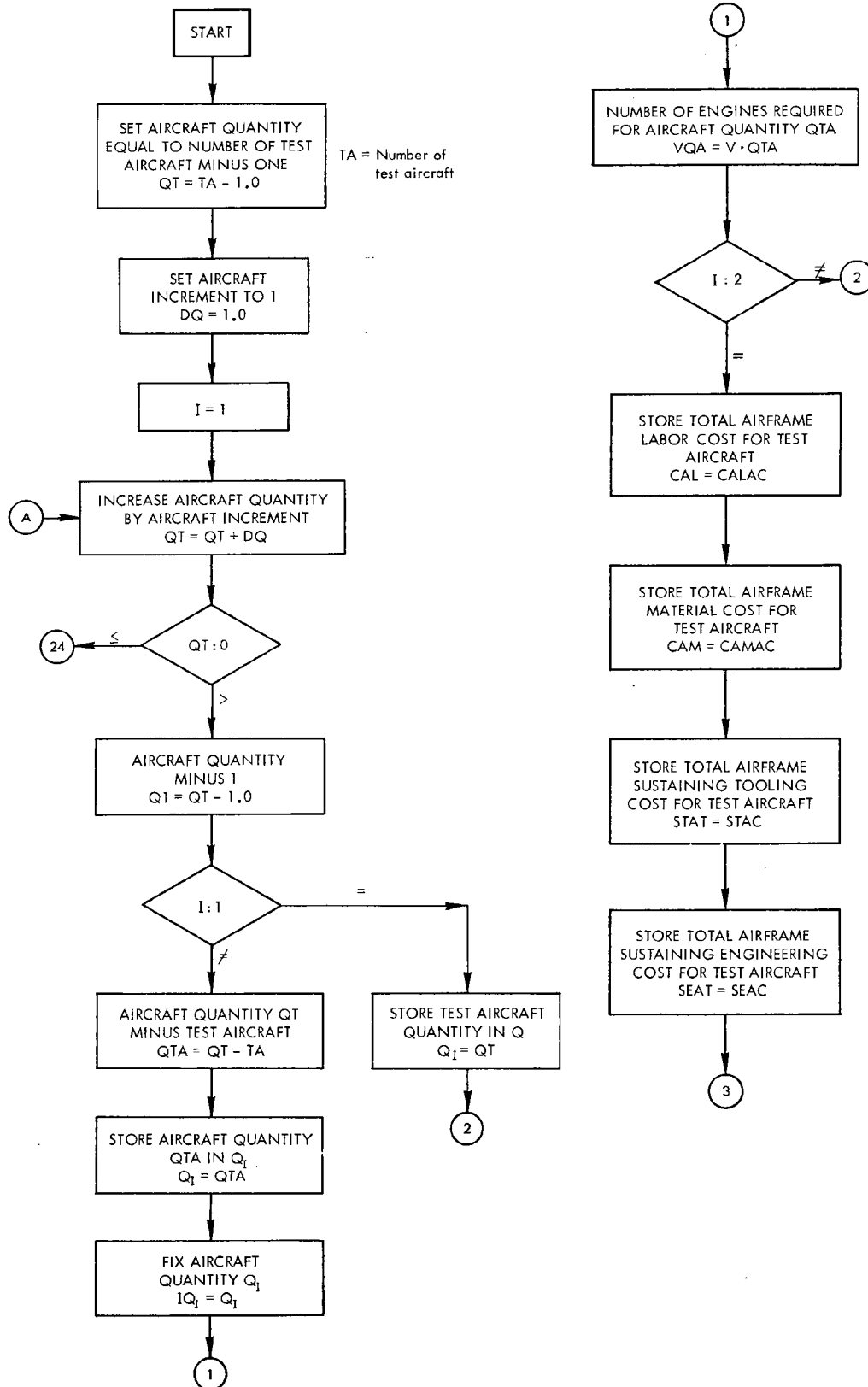


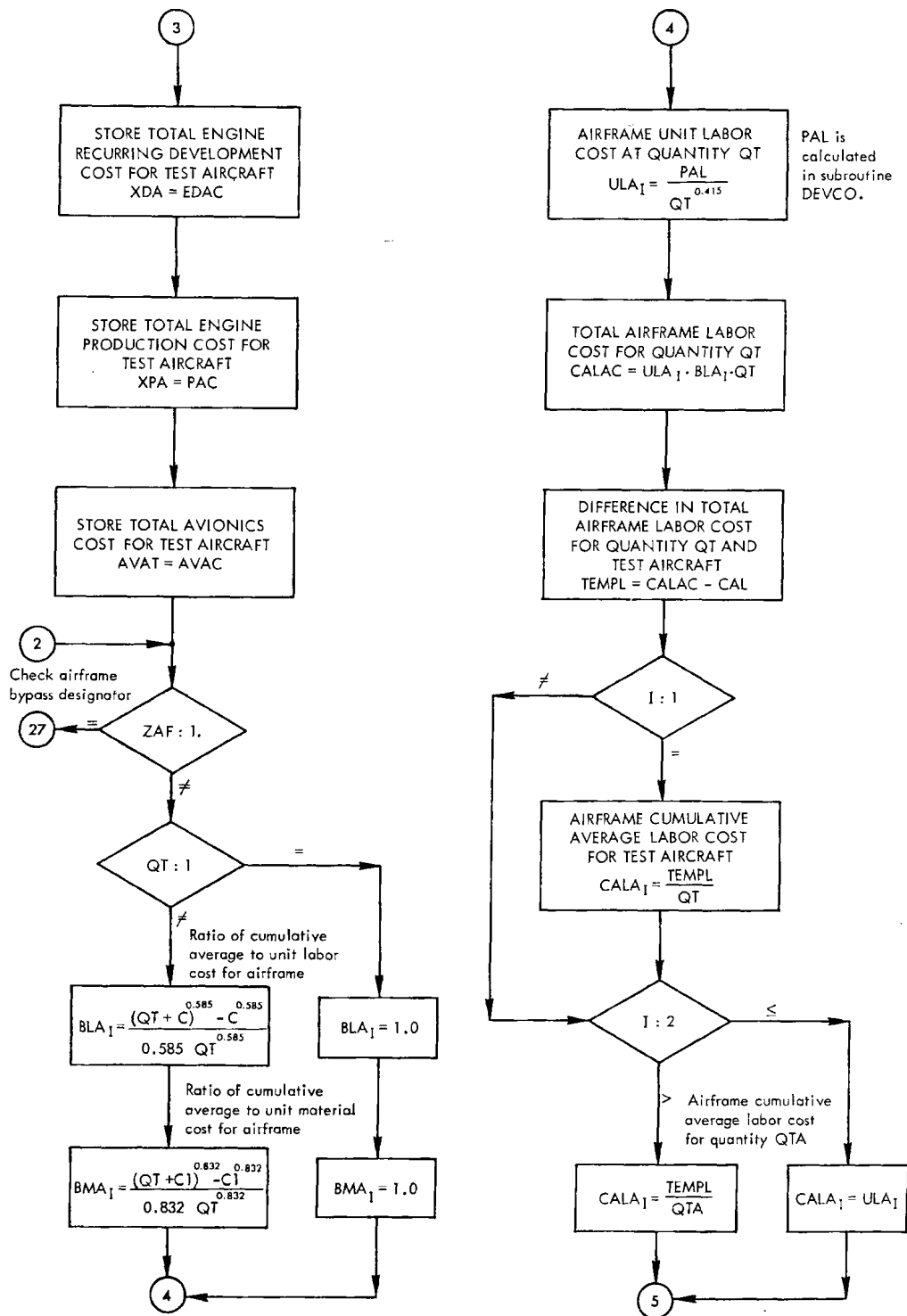
*All costs are calculated in millions of dollars.

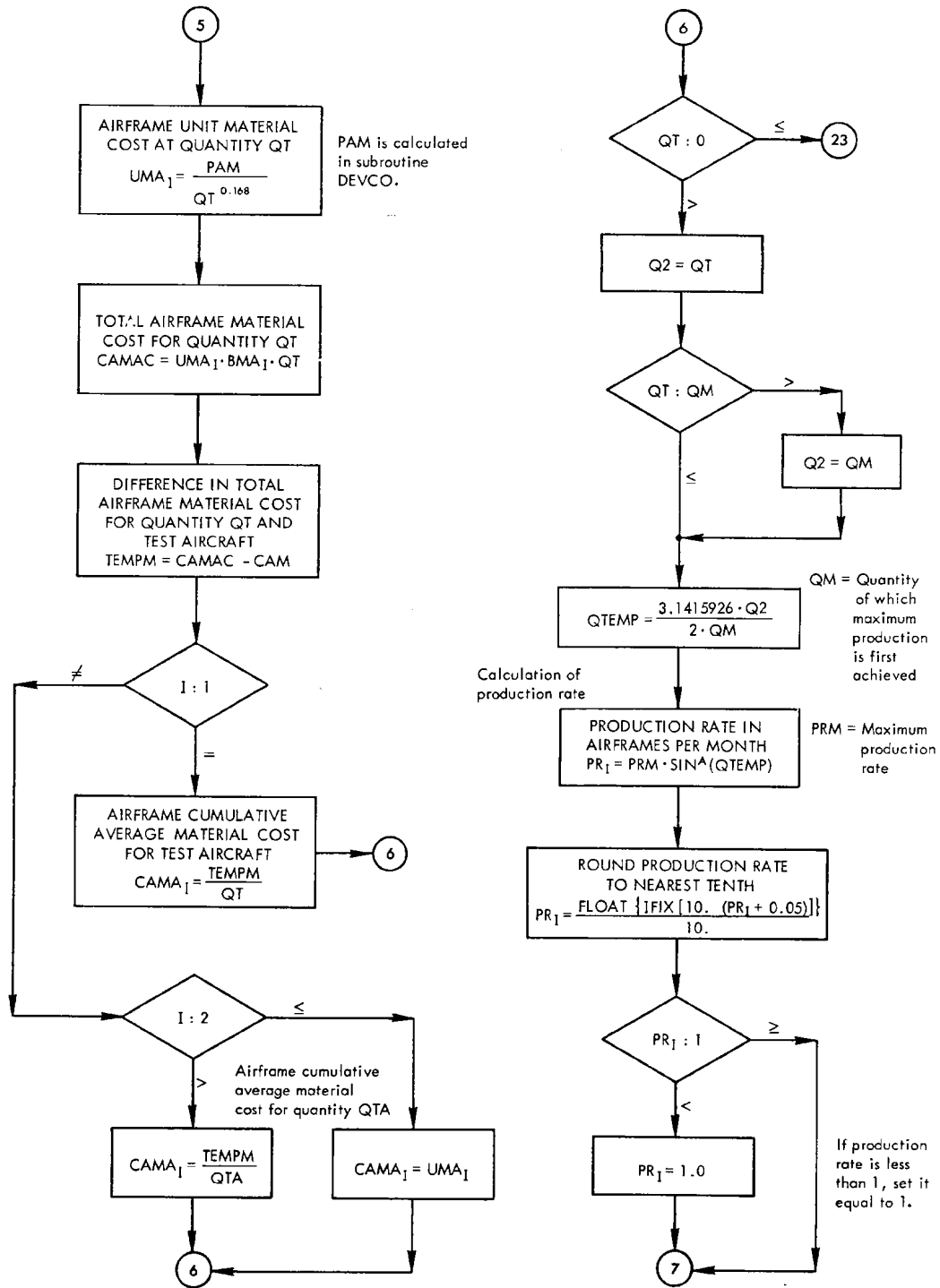


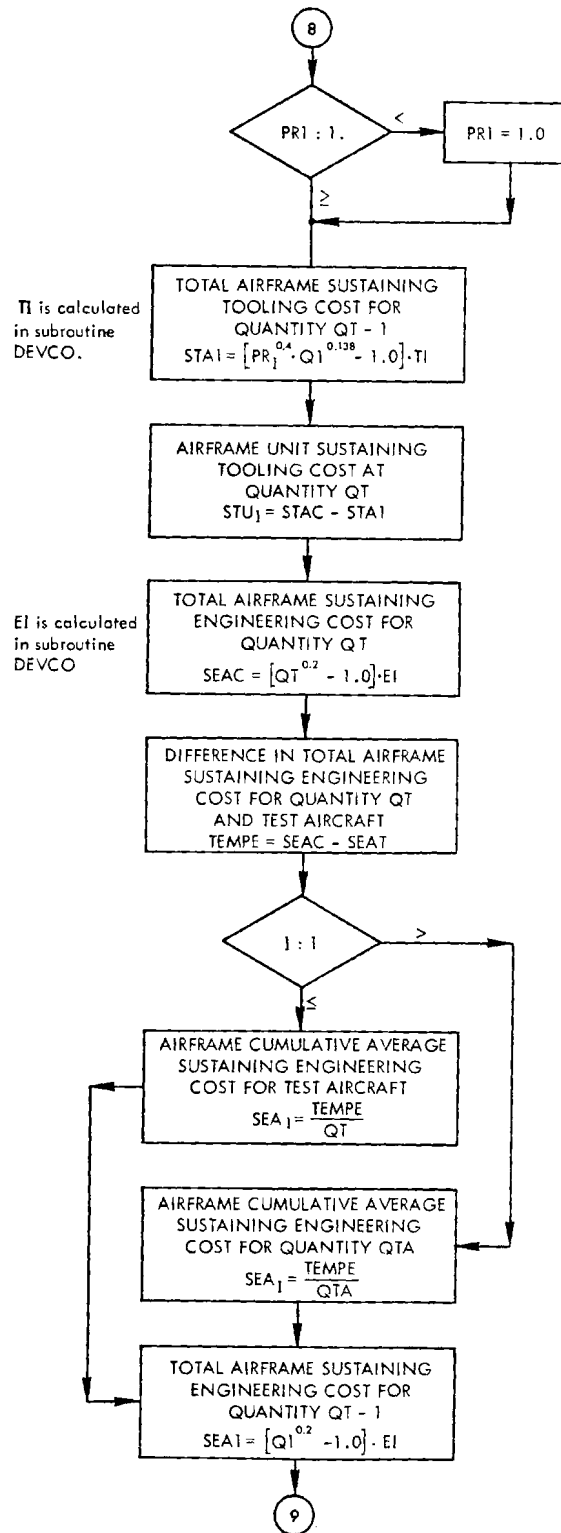
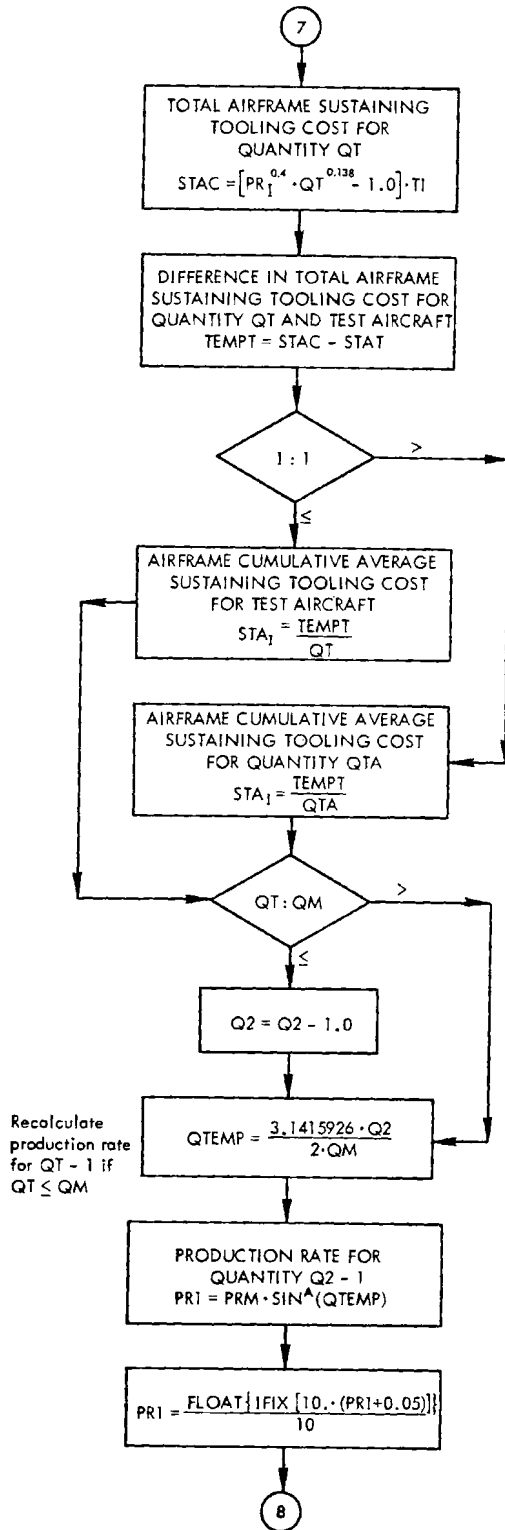


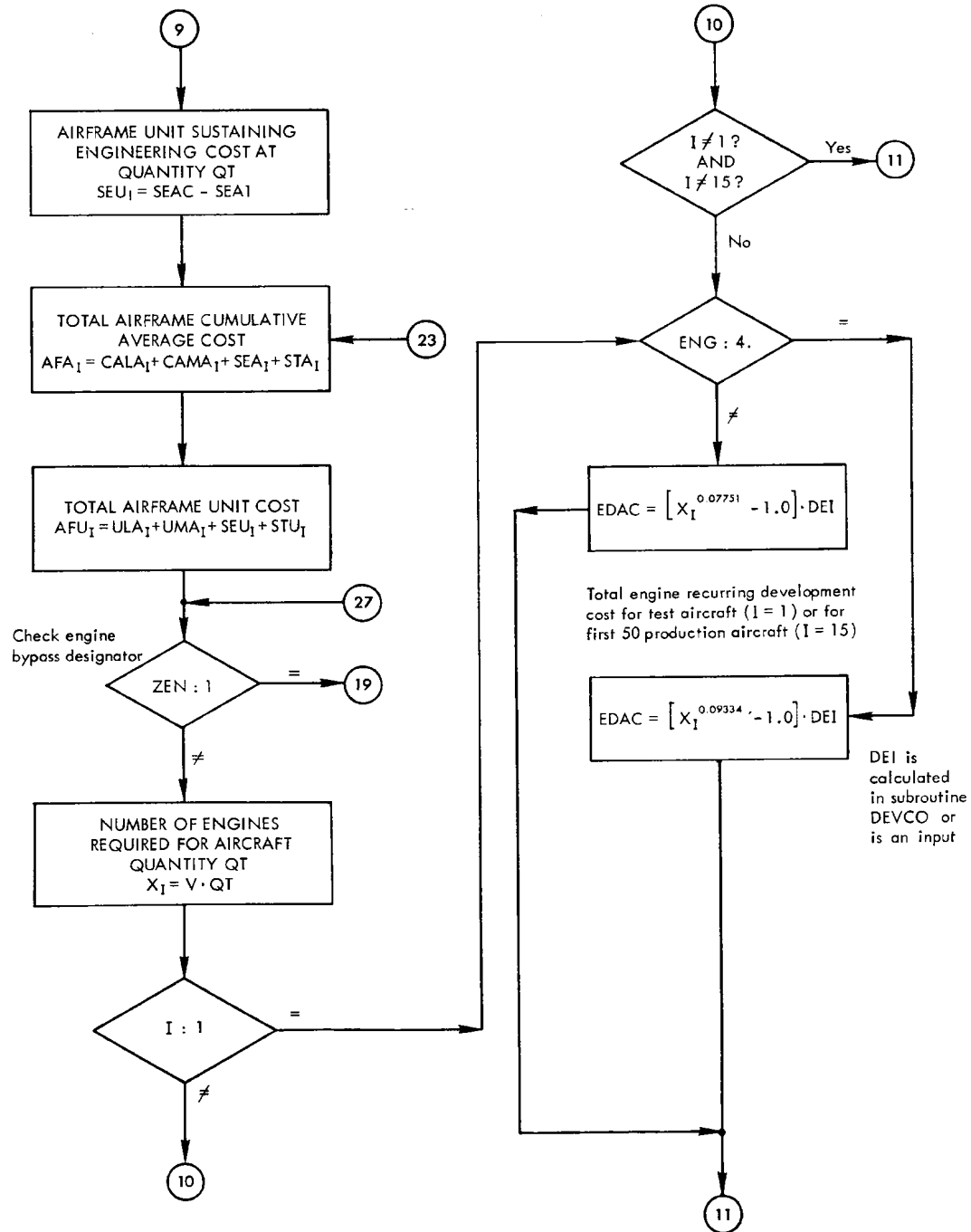
Subroutine PRODCO

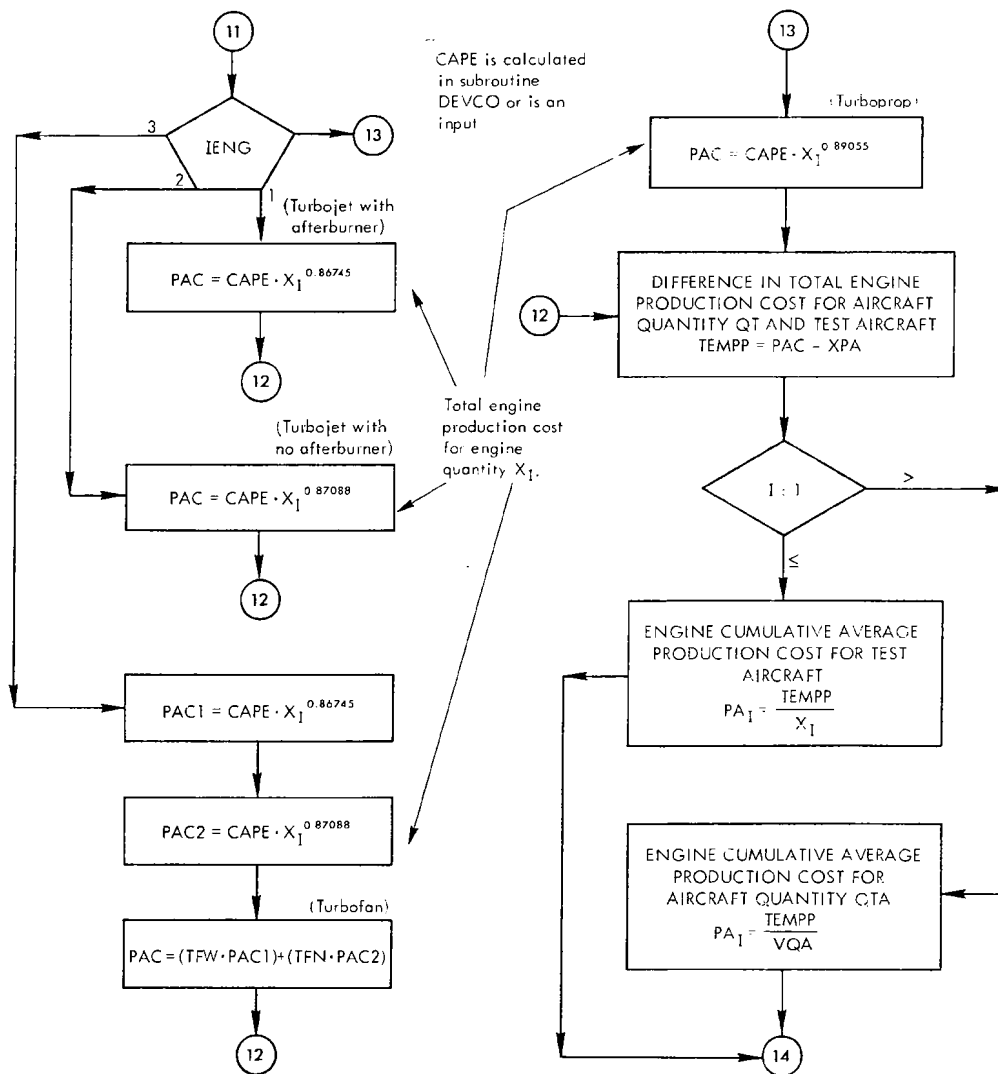


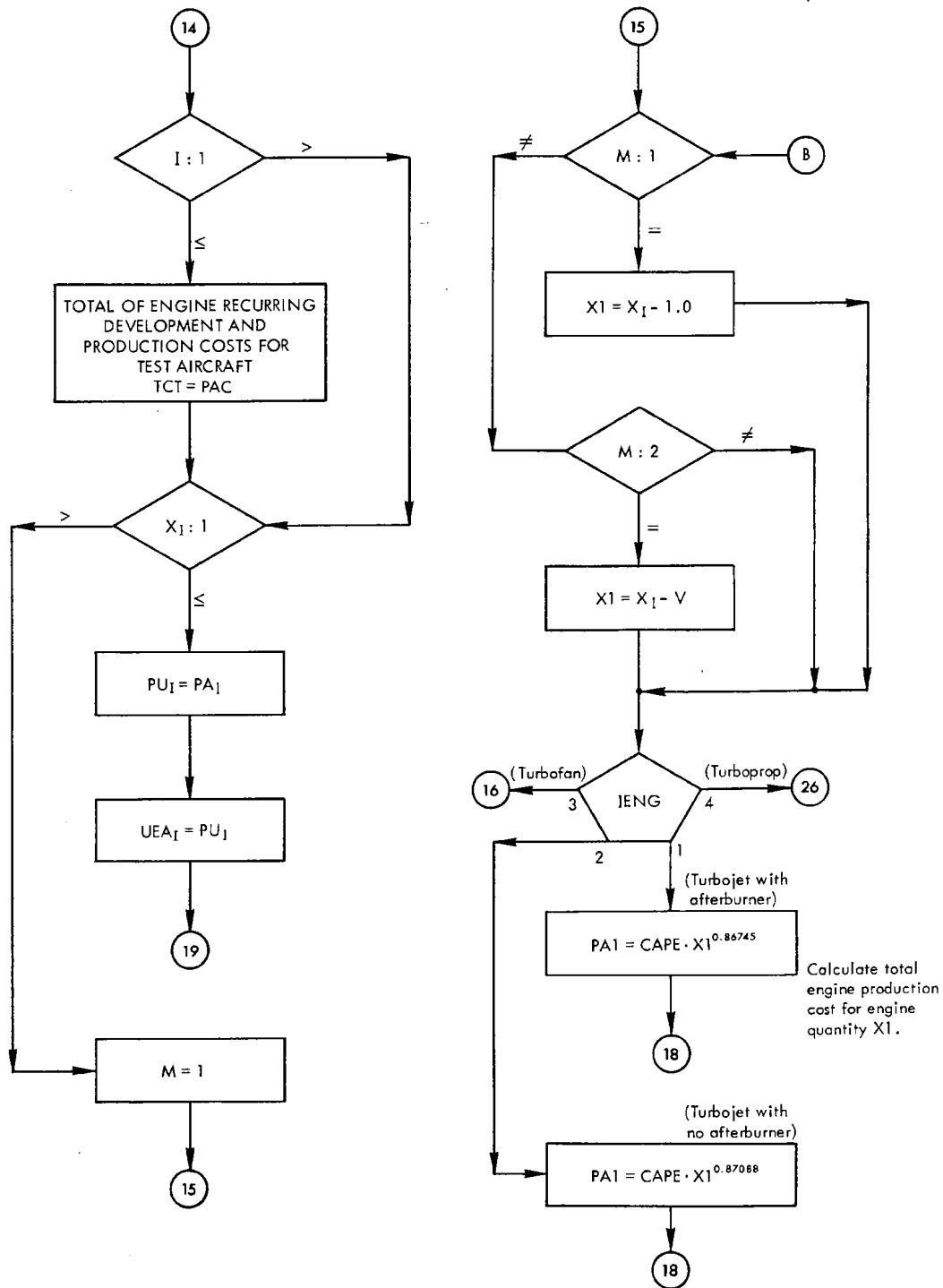


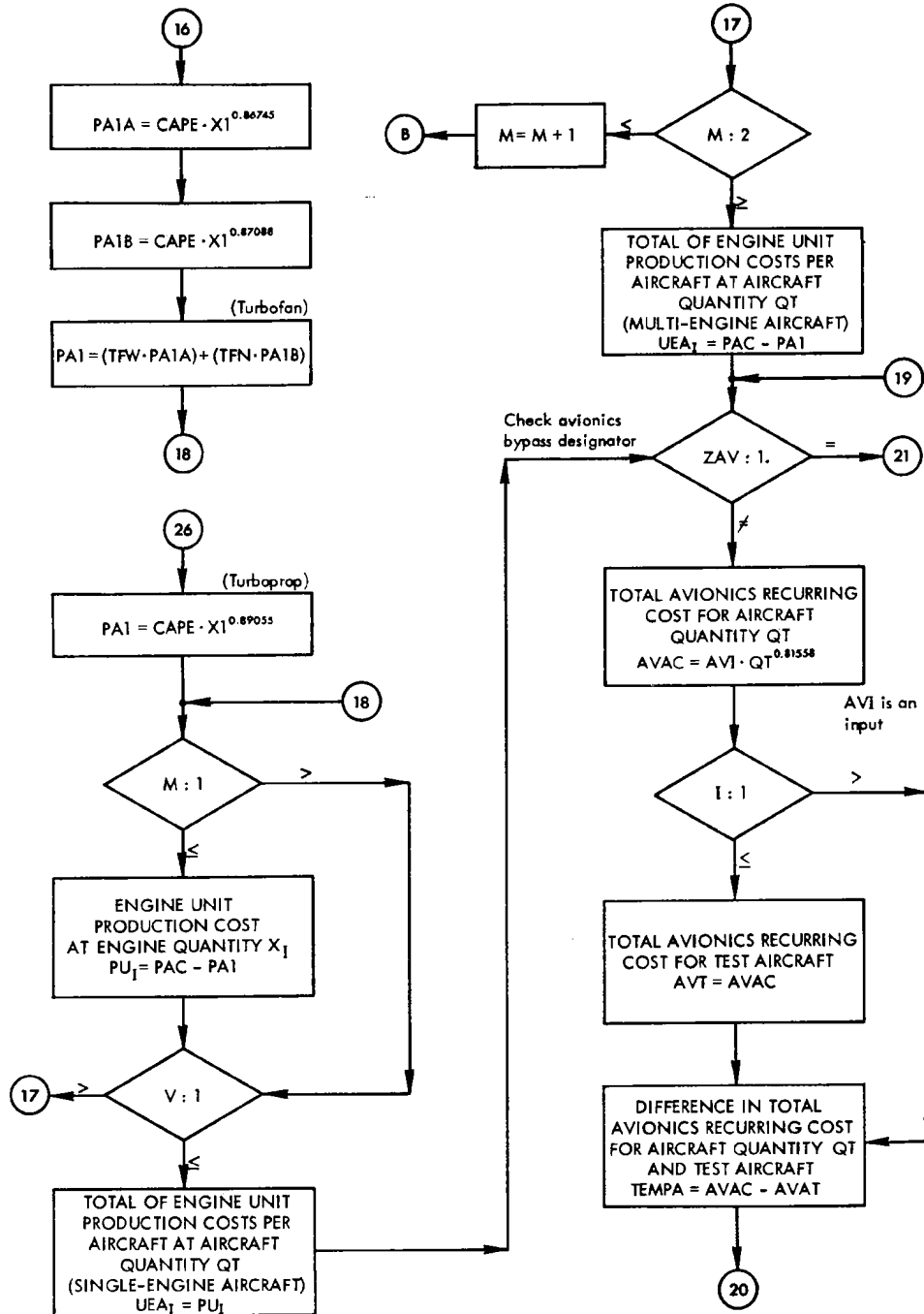


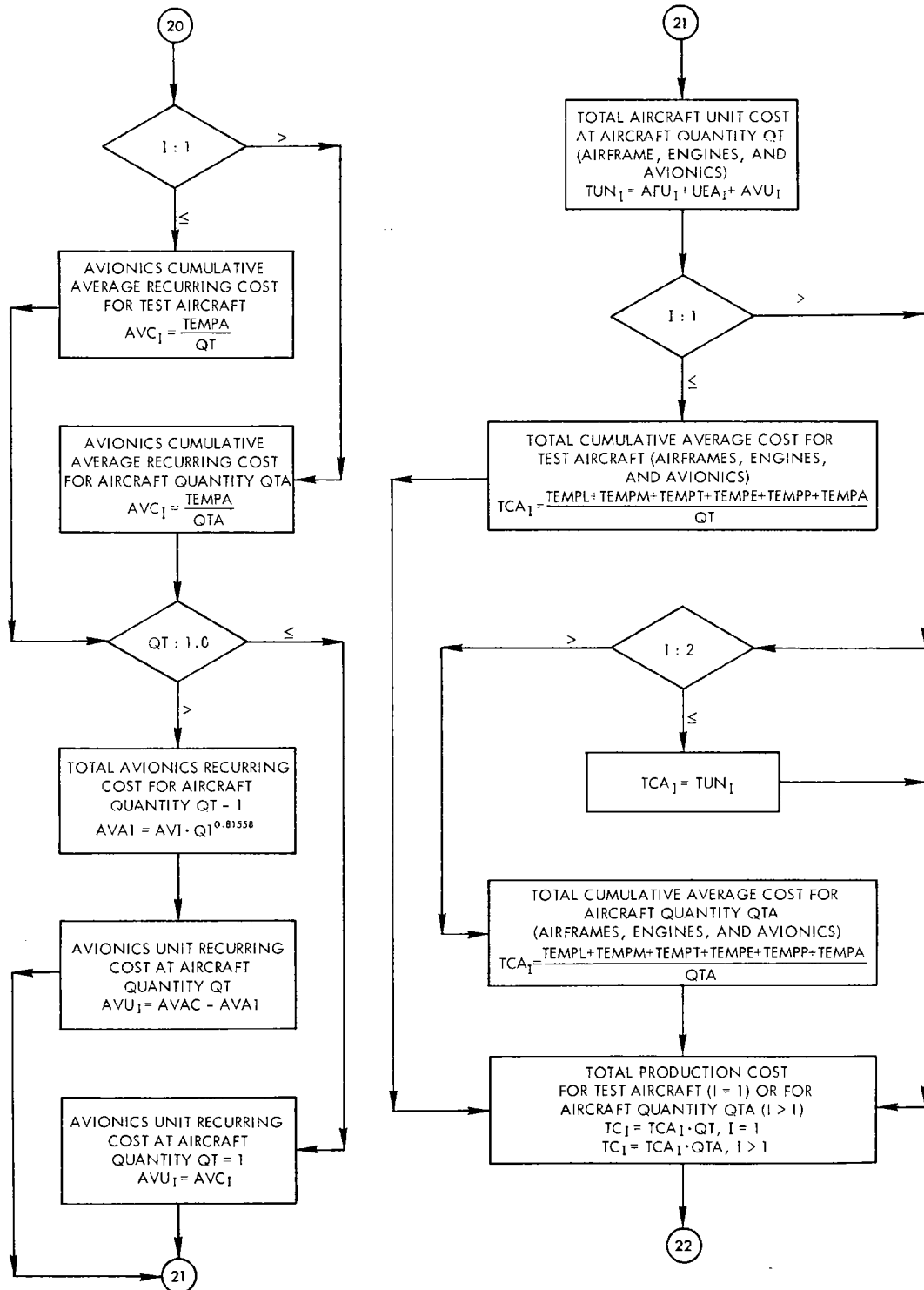


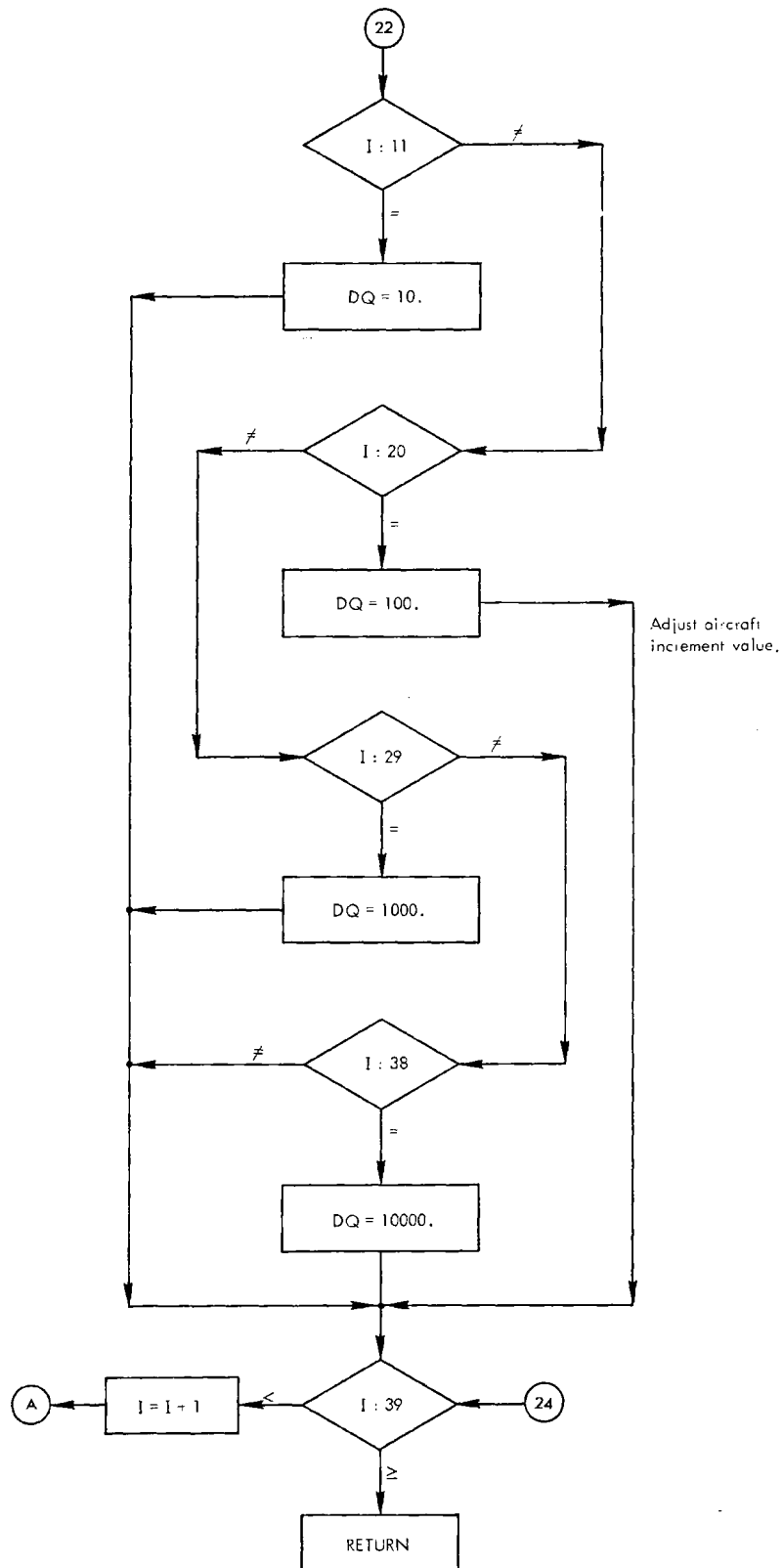




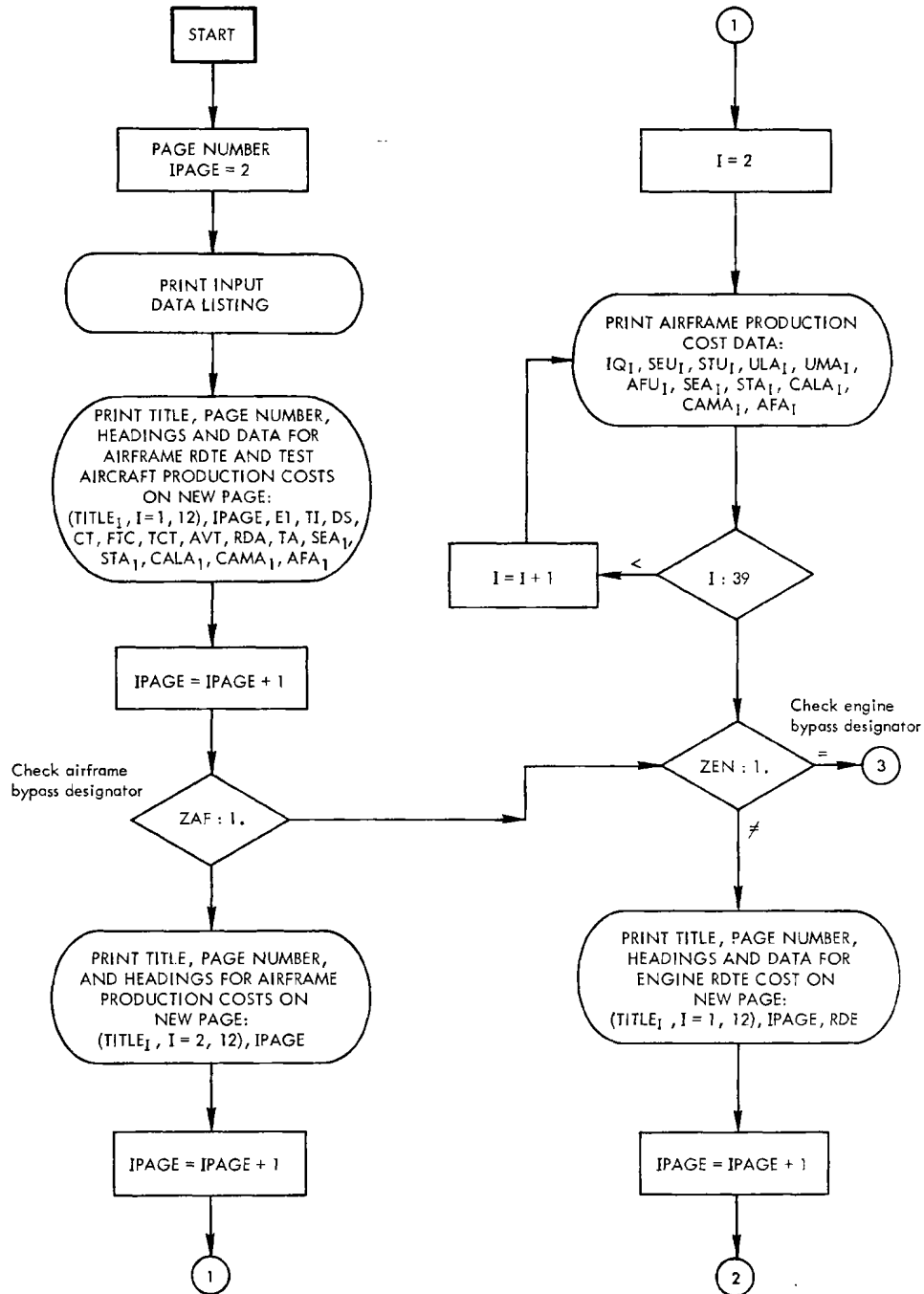


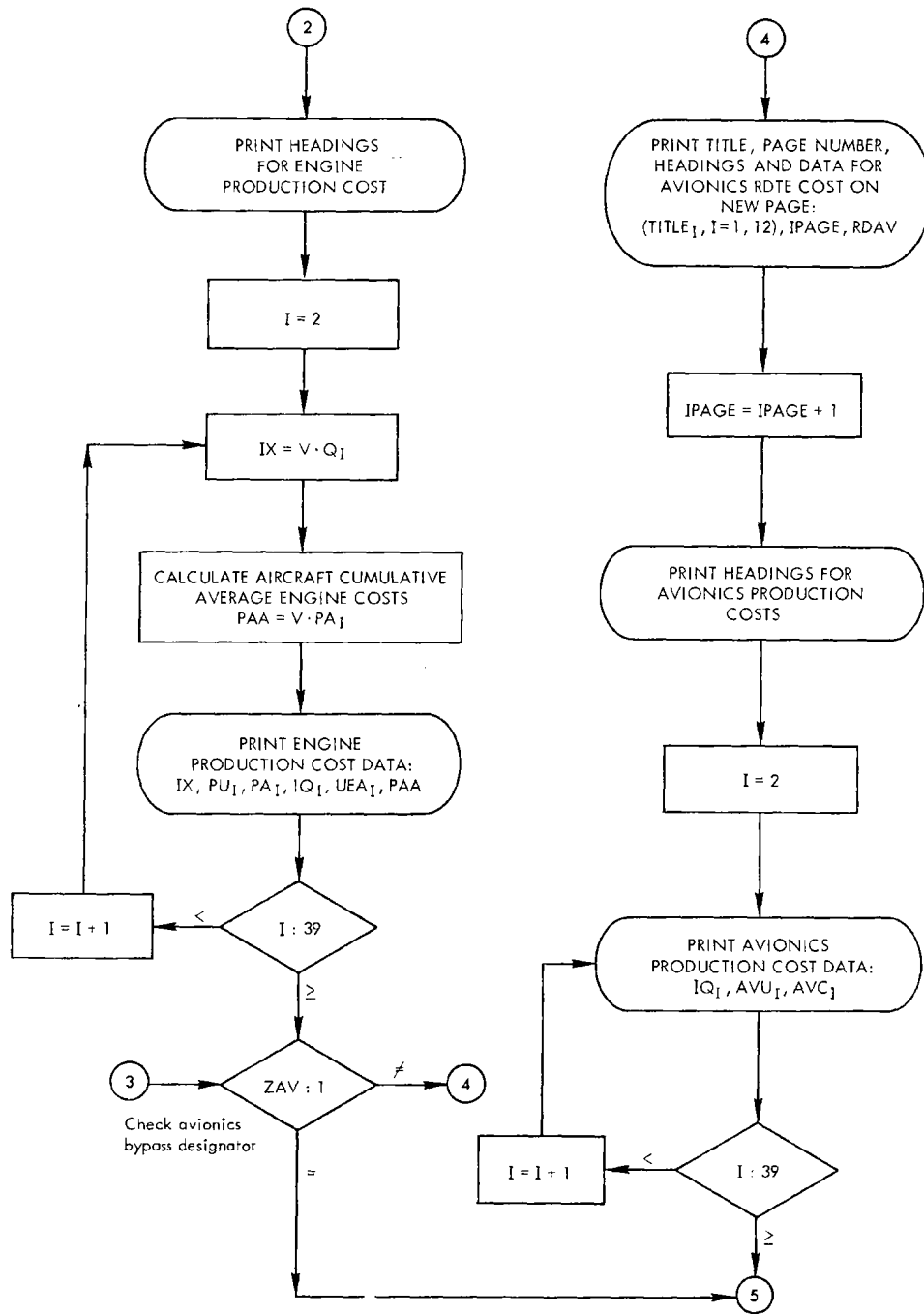


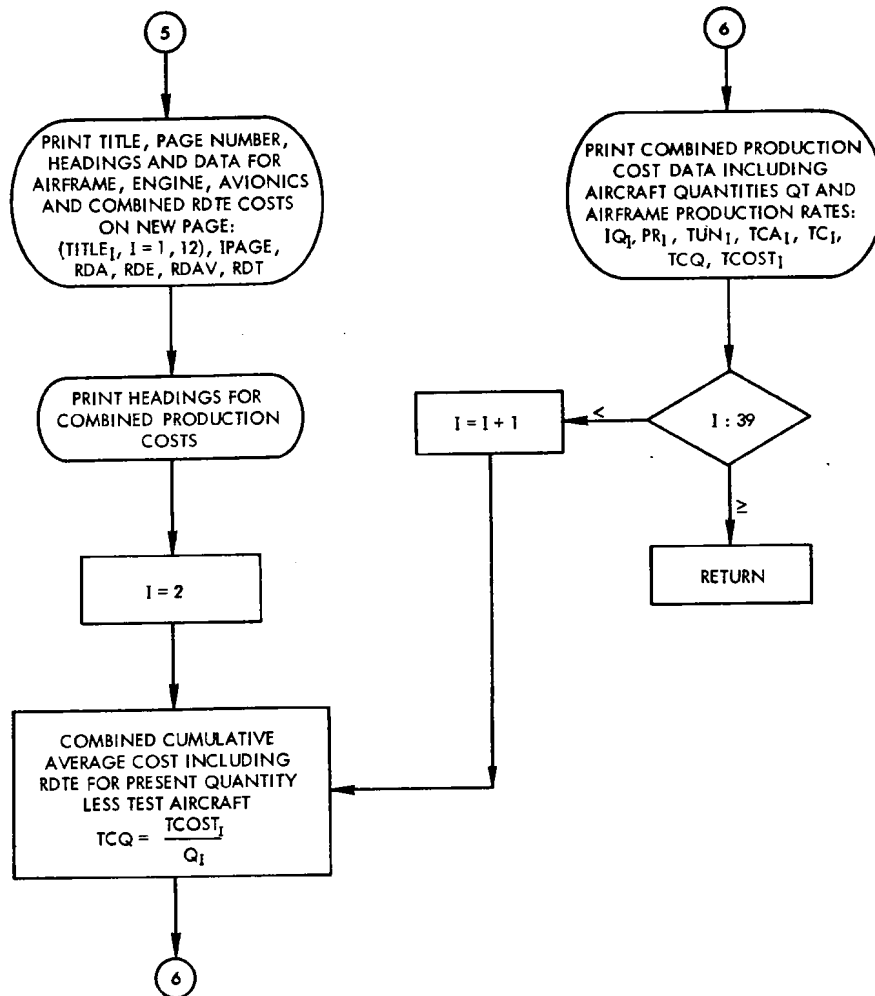




Subroutine OUTPUT







Appendix C

LISTING OF FORTRAN-IV SOURCE PROGRAM

\$IBFTC MAIN

```
COMMON RUN, W, S, PRM, QM, TA, ENG, T, V, RDAV, AVI, ETIN, DEI,
C CAPE, A, ZAF, ZEN, ZAV, P, CR1, CR2, CR3, CR4, CR5, CR6, TFW,
C CLEAR
COMMON AVAC, AVAT, AVA1, AVT, C, C1, CAL, CALAC, CAM, CAMAC, CT,
C CTE, DS, EDAC, EDAL, EI, FTC, PAC, PAL, PAM, PA1, QT, QTA, Q1,
C Q2, RDA, RDE, RDT, SEAC, SEAT, SEAL, STAC, STAT, STA1, TCT,
C TEMPA, TEMPE, TEMPL, TEMPM, TEMPP, TEMPT, TI, TT, VQA, XCAPE,
C XDA, XPA, X1, INDT, IENG, INDE, TFN, IRUN, IPAGE
COMMON AFA, AFU, AVC, AVU, BLA, BMA, CALA, CAMA, PA, PR, PU, Q,
C SEA, SEU, STA, STU, TC, TCA, TCUST, TUN, UEA, ULA, UMA, X, IQ,
C GDATA, IDATA, TITLE
DIMENSION AFA(40), AFU(40), AVC(40), AVU(40), BLA(40), BMA(40),
C CALA(40), CAMA(40), PA(40), PR(40), PU(40), Q(40), SEA(40),
C SEU(40), STA(40), STU(40), TC(40), TCA(40), TCOST(40), TUN(40),
C UEA(40), ULA(40), UMA(40), X(40), IQ(40), GDATA(100), IDATA(100),
C TITLE(12), F(1)
EQUIVALENCE (F(1), RUN), (T, ESH)
```

```
C
C MAIN ROUTINE OF COMPUTER PROGRAM (DAPCA) THAT DETERMINES RDTE AND
C PRODUCTION (RECURRING) COSTS FOR AIRFRAMES, ENGINES, AND AVIONICS
C (AVIONICS RDTE COST IS A THRUPUT)
C
C SET TITLE DESIGNATOR TO 1
10 INDT = 1
C CLEAR ALL STORAGE LOCATIONS FOR INPUT DATA
20 DO 30 I = 1, 27
  F(I) = 0.
30 CONTINUE
C CLEAR REMAINING STORAGE LOCATIONS IN COMMON EXCEPT FOR TITLE
C DESIGNATOR AND TITLE
40 DO 50 I = 28, 1281
  IF (I .EQ. 76) GO TO 50
  F(I) = 0.
50 CONTINUE
60 CALL READ
  CALL DEVCO
  CALL OUTPUT
  IF (CLEAR .EQ. 1.) GO TO 20
  GO TO 40
END
```


\$IBFTC READ

SUBROUTINE READ

COMMON RUN, W, S, PRM, QM, TA, ENG, T, V, RDAV, AVI, ETIN, DEI,
 C CAPE, A, ZAF, ZEN, ZAV, P, CR1, CR2, CR3, CR4, CR5, CR6, TFW,
 C CLEAR
 COMMON AVAC, AVAT, AVA1, AVT, C, C1, CAL, CALAC, CAM, CAMAC, CT,
 C CTE, DS, EDAC, EDA1, EI, FTC, PAC, PAL, PAM, PA1, QT, QTA, Q1,
 C Q2, RDA, RDE, RDT, SEAC, SEAT, SEA1, STAC, STAT, STA1, TCT,
 C TEMPA, TEMPE, TEMPL, TEMPM, TEMPP, TEMPT, TI, TT, VQA, XCAPE,
 C XDA, XPA, X1, INDT, IENG, INDE, TFN, IRUN, IPAGE
 COMMON AFA, AFU, AVC, AVU, BLA, BMA, CALA, CAMA, PA, PR, PU, Q,
 C SEA, SEU, STA, STU, TC, TCA, TCOST, TUN, UEA, ULA, UMA, X, IQ,
 C GDATA, IDATA, TITLE
 DIMENSION AFA(40), AFU(40), AVC(40), AVU(40), BLA(40), BMA(40),
 C CALA(40), CAMA(40), PA(40), PR(40), PU(40), Q(40), SEA(40),
 C SEU(40), STA(40), STU(40), TC(40), TCA(40), TCOST(40), TUN(40),
 C UEA(40), ULA(40), UMA(40), X(40), IQ(40), GDATA(100), IDATA(100),
 C TITLE(12), F(1)
 EQUIVALENCE (F(1), RUN), (T, ESH)

C
 C SUBROUTINE FOR READING IN INPUT DATA
 C
 C SET PAGE NUMBER TO 1
 10 IPAGE = 1
 C READ NEW TITLE IF SO INDICATED (IND = 1) -- STARTING IN COLUMN 1
 C AND, AS REQUIRED, ENDING IN COLUMN 72
 IF (INDT .EQ. 1) READ (5, 20) (TITLE(I), I = 1, 12)
 20 FORMAT (12A6)
 C SET TITLE DESIGNATOR TO ZERO
 INDT = 0
 C READ INPUT DATA
 K = 1
 DO 50 L = 1, 30
 K1 = K + 2
 READ (5, 30) ((IDATA(J), GDATA(J)), J = K, K1)
 30 FORMAT (3(I5, F19.9))
 DO 40 J = K, K1
 IF (IDATA(J) .EQ. 5555) GO TO 60
 IF (IDATA(J) .EQ. 9999) GO TO 70
 40 CONTINUE
 K = K + 3
 50 CONTINUE
 C WRITE ERROR MESSAGE AND TERMINATE THIS AND REMAINING RUNS.
 WRITE (6, 52) (TITLE(L), L = 1, 12)
 52 FORMAT (1H1, 9X, 12A6 ///// 9X, 92HNUMBER OF INPUT DATA CARDS HAS
 C EXCEEDED MAXIMUM ALLOWABLE (30) FOR THIS RUN. JOB TERMINATED.)
 CALL EXIT
 60 INDT = 1
 C STORE INPUT DATA ACCORDING TO CELL ADDRESSES
 70 DO 80 I = 1, 100
 IF (IDATA(I) .EQ. 5555 .OR. IDATA(I) .EQ. 9999) GO TO 90
 IF (IDATA(I) .EQ. 0 .AND. GDATA(I) .EQ. 0.) GO TO 80
 IF (IDATA(I) .EQ. 0 .AND. GDATA(I) .NE. 0.) GO TO 82
 IF (IDATA(I) .LT. 0 .OR. IDATA(I) .GT. 27) GO TO 82
 J = IDATA(I)
 F(J) = GDATA(I)

```
80 CONTINUE
   GO TO 90
C   WRITE ERROR MESSAGE AND TERMINATE THIS AND REMAINING RUNS.
82 WRITE (6, 84) (TITLE(L), L = 1, 12)
84 FORMAT (1H1, 9X, 12A6 ///// 9X, 79HAN ADDRESS FOR AN INPUT VARIA
   CBLE HAS NOT BEEN ENTERED PROPERLY. JOB TERMINATED.)
   CALL EXIT
C   FIX RUN NUMBER, ENGINE TYPE DESIGNATOR, AND ENGINE THRUPUT
C   DESIGNATOR
90 IRUN = RUN
   IENG = ENG
   INDE = ETIN
C   PRINT TITLE, PAGE NUMBER, AND RUN NUMBER ON NEW PAGE
   WRITE (6, 100) (TITLE(I), I = 1, 12), IPAGE, IRUN
100 FORMAT (1H1, 9X, 12A6, 38X, 5HPAGE , 12 / 10X, 4HRUN , 14)
C   STEP PAGE NUMBER BY 1
   IPAGE = IPAGE + 1
C   SET EXPONENT A TO 2 IF NOT ENTERED
   IF (A .EQ. 0.) A = 2
C   SET WEIGHTING FACTOR FOR TURBOJET WITH AFTERBURNER TO 0.5 IF NOT
C   ENTERED (USED IN TURBOFAN CALCULATIONS)
   IF (TFW .EQ. 0.) TFW = 0.5
C   WEIGHTING FACTOR FOR TURBOJET WITH NO AFTERBURNER
   TFN = 1.0 - TFW
C   SET AIRFRAME PROFIT FACTOR TO 0.1 AND ADJUSTMENT FACTORS TO 1 IF
C   NOT ENTERED
   DO 110 I = 19, 25
   IF (I .EQ. 19 .AND. F(I) .EQ. 0.) F(I) = 0.1
   IF (I .GT. 19 .AND. F(I) .EQ. 0.) F(I) = 1.
110 CONTINUE
C   ZERO OUT POSSIBLE THRUPUT COSTS RETAINED FROM PREVIOUS RUN IF
C   THEIR RESPECTIVE CATEGORIES ARE TO BE BYPASSED IN THIS RUN.
   IF (ZEN .NE. 1.) GO TO 120
   DEI = 0.
   CAPE = 0.
120 IF (ZAV .NE. 1.) RETURN
   RDAV = 0.
   AVI = 0.
   RETURN
   END
```

\$IBFTC DEVCO

SUBROUTINE DEVCO

COMMON RUN, W, S, PRM, QM, TA, ENG, T, V, RDAV, AVI, ETIN, DEI,
C CAPE, A, ZAF, ZEN, ZAV, P, CR1, CR2, CR3, CR4, CR5, CR6, TFW,
C CLEAR

COMMON AVAC, AVAT, AVA1, AVT, C, C1, CAL, CALAC, CAM, CAMAC, CT,
C CTE, DS, EDAC, EDA1, EI, FTC, PAC, PAL, PAM, PA1, QT, QTA, Q1,
C Q2, RDA, RDE, RDT, SEAC, SEAT, SEA1, STAC, STAT, STA1, TCT,
C TEMPA, TEMPE, TEMPL, TEMPM, TEMPP, TEMPT, TI, TT, VQA, XCAPE,
C XDA, XPA, X1, INDT, IENG, INDE, TFN, IRUN, IPAGE

COMMON AFA, AFU, AVC, AVU, BLA, BMA, CALA, CAMA, PA, PR, PU, Q,
C SEA, SEU, STA, STU, TC, TCA, TCOST, TUN, UEA, ULA, UMA, X, IQ,
C GDATA, IDATA, TITLE

DIMENSION AFA(40), AFU(40), AVC(40), AVU(40), BLA(40), BMA(40),
C CALA(40), CAMA(40), PA(40), PR(40), PU(40), Q(40), SEA(40),
C SEU(40), STA(40), STU(40), TC(40), TCA(40), TCOST(40), TUN(40),
C UEA(40), ULA(40), UMA(40), X(40), IQ(40), GDATA(100), IDATA(100),
C TITLE(12), F(1)

EQUIVALENCE (F(1), RUN), (T, ESH)

SUBROUTINE FOR DETERMINING INITIAL COSTS, TOTAL DEVELOPMENT COSTS,
AND GRAND TOTAL COSTS

10 DM = 0.000001 * (1.0 + P)
IF (ENG .NE. 4.) TT = T * V
IF (ENG .EQ. 4.) TT = ESH * V * 2.5
IF (ZAF .EQ. 1.) GO TO 15
C INITIAL ENGINEERING COST FOR AIRFRAME (IN MILLIONS OF DOLLARS)
EI = (10.0**0.90462) * (S**0.54716) * (TT**0.88) * 12.25 * DM * CR1
C DEVELOPMENT SUPPORT COST FOR AIRFRAME (IN MILLIONS OF DOLLARS)
DS = 15.6 * EI/12.25
C INITIAL TOOLING COST FOR AIRFRAME (IN MILLIONS OF DOLLARS)
TI = ((S**1.07437) * (W**0.83913)/(10.0**0.91248)) * 10.55 * DM * CR2
C STARTING POSITION FOR CALCULATING RATIO OF CUMULATIVE AVERAGE
C TO UNIT COST FOR AIRFRAME LABOR
C = 0.5
C STARTING POSITION FOR CALCULATING RATIO OF CUMULATIVE AVERAGE
C TO UNIT COST FOR AIRFRAME MATERIAL
C1 = 0.5
C UNIT LABOR COST FOR AIRFRAME NUMBER 1
PAL = 6.762 * (10.0**0.16314) * (W**0.73672) * (S**0.43113) * 9.50
C * 1.11 * 1.14 * DM * CR3
C UNIT MATERIAL COST FOR AIRFRAME NUMBER 1
PAM = (2.169 * (W**0.77933) * (S**0.85650)/(10.0**0.76558)) * DM * CR4
15 IF (ZEN .EQ. 1.) GO TO 75
IF (INDE .EQ. 3) GO TO 70
IF (INDE .EQ. 1) GO TO 20
C INITIAL DEVELOPMENT COST FOR ENGINES
IF (ENG .NE. 4.) DEI = 0.13937 * (T**0.74356) * CR5
IF (ENG .EQ. 4.) DEI = 2.82917 * (ESH**0.35497) * CR5
20 IF (INDE .EQ. 2) GO TO 70
C PROCUREMENT COST FOR FIRST PRODUCTION-LINE ENGINE
GO TO (30, 40, 50, 60), IENG
30 CAPE = 0.18700 * (T**0.84845) * CR6
GO TO 70
40 CAPE = 0.31979 * (T**0.81626) * CR6

```
GO TO 70
50 CAPE1 = 0.18700 * (T**0.84845) * CR6
   CAPE2 = 0.31979 * (T**0.81626) * CR6
   CAPE = (TFW * CAPE1) + (TFN * CAPE2)
   GO TO 70
60 CAPE = 4.86224 * (ESH**0.45873) * CR6
C  STORE VALUE OF CAPE
70 XCAPE = CAPE
   CAPE = .001 * XCAPE
75 IF (ZAF .EQ. 1.) GO TO 77
C  FLIGHT TEST COST
   FTC = 0.58 * (TA**1.1) * (W**0.80) * (S**0.90) * DM
77 CALL PRODCO
C  TOTAL DEVELOPMENT COST FOR ENGINES (INITIAL DEVELOPMENT COST PLUS
C  RECURRING DEVELOPMENT COST FOR ENGINES FOR FIRST 50 PRODUCTION
C  AIRCRAFT
   IF (INDE .EQ. 0 .OR. INDE .EQ. 2) DEI = DEI + EDAC
C  TOTAL AIRFRAME COST OF TEST AIRCRAFT
80 CT = TA * AFA(1)
C  TOTAL AIRFRAME COST OF TEST AIRCRAFT INCLUDING ENGINES AND
C  AVIONICS FOR TEST AIRCRAFT
   CTE = CT + TCT + AVT
C  TOTAL RESEARCH AND DEVELOPMENT COST FOR AIRFRAME
   RDA = EI + TI + DS + FTC + CTE
C  TOTAL RESEARCH AND DEVELOPMENT COST FOR ENGINES
   RDE = DEI
C  TOTAL RESEARCH AND DEVELOPMENT COST FOR AIRCRAFT (AIRFRAME,
C  ENGINES, AND AVIONICS)
   RDT = RDA + RDE + RDAV
C  GRAND TOTAL COST OF AIRFRAMES, ENGINES, AND AVIONICS FOR
C  QUANTITIES USED ON AIRFRAME COST CURVE (PRODUCTION COSTS PLUS
C  RESEARCH AND DEVELOPMENT COSTS)
   DO 90 I = 2, 39
   TCOST(I) = TC(I) + RDT
90 CONTINUE
C  RESTORE CAPE TO ITS INITIAL VALUE
   CAPE = XCAPE
   RETURN
END
```

\$IBFTC PRODCO

SUBROUTINE PRODCO

COMMON RUN, W, S, PRM, QM, TA, ENG, T, V, RDAV, AVI, ETIN, DEI,
C CAPE, A, ZAF, ZEN, ZAV, P, CR1, CR2, CR3, CR4, CR5, CR6, TFW,
C CLEAR

COMMON AVAC, AVAT, AVAL, AVT, C, C1, CAL, CALAC, CAM, CAMAC, CT,
C CTE, DS, EDAC, EDA1, EI, FTC, PAC, PAL, PAM, PA1, QT, QTA, Q1,
C Q2, RDA, RDE, RDT, SEAC, SEAT, SEA1, STAC, STAT, STA1, TCT,
C TEMPA, TEMPE, TEMPL, TEMPM, TEMPP, TEMPT, T1, TT, VQA, XCAPE,
C XDA, XPA, X1, INDT, IENG, INDE, TFN, IRUN, IPAGE

COMMON AFA, AFU, AVC, AVU, BLA, BMA, CALA, CAMA, PA, PR, PU, Q,
C SEA, SEU, STA, STU, TC, TCA, TCOST, TUN, UEA, ULA, UMA, X, IQ,
C GDATA, IDATA, TITLE

DIMENSION AFA(40), AFU(40), AVC(40), AVU(40), BLA(40), BMA(40),
C CALA(40), CAMA(40), PA(40), PR(40), PU(40), Q(40), SEA(40),
C SEU(40), STA(40), STU(40), TC(40), TCA(40), TCOST(40), TUN(40),
C UEA(40), ULA(40), UMA(40), X(40), IQ(40), GDATA(100), IDATA(100),
C TITLE(12), F(1)

EQUIVALENCE (F(1), RUN), (T, ESH)

C
C SUBROUTINE FOR DETERMINING PRODUCTION (RECURRING) COSTS

C SET QUANTITY OF AIRCRAFT EQUAL TO TEST AIRCRAFT MINUS 1

10 QT = TA - 1.0

C SET AIRCRAFT INCREMENT TO 1

DQ = 1.0

DO 210 I = 1, 39

C INCREASE AIRCRAFT QUANTITY BY AIRCRAFT INCREMENT

QT = QT + DQ

IF (QT .LE. 0.) GO TO 210

C AIRCRAFT QUANTITY QT-1

Q1 = QT - 1.0

C STORE TEST AIRCRAFT QUANTITY IN Q(1)

IF (I .EQ. 1) Q(1) = QT

IF (I .EQ. 1) GO TO 30

C AIRCRAFT QUANTITY QT MINUS TEST AIRCRAFT

QTA = QT - TA

C STORE AIRCRAFT QUANTITY QTA IN Q(I) (I GREATER THAN 1)

Q(I) = QTA

C FIX AIRCRAFT QUANTITY Q(I)

IQ(I) = Q(I)

C NUMBER OF ENGINES REQUIRED FOR AIRCRAFT QUANTITY QTA (V = NUMBER

C OF ENGINES PER AIRCRAFT)

VQA = V * QTA

20 IF (I .NE. 2) GO TO 30

C STORE TOTAL AIRFRAME LABOR COST FOR TEST AIRCRAFT

CAL = CALAC

C STORE TOTAL AIRFRAME MATERIAL COST FOR TEST AIRCRAFT

CAM = CAMAC

C STORE TOTAL AIRFRAME SUSTAINING TOOLING COST FOR TEST AIRCRAFT

STAT = STAC

C STORE TOTAL AIRFRAME SUSTAINING ENGINEERING COST FOR TEST AIRCRAFT

SEAT = SEAC

C STORE TOTAL ENGINE RECURRING DEVELOPMENT COST FOR TEST AIRCRAFT

XDA = EDAC

C STORE TOTAL ENGINE PRODUCTION COST FOR TEST AIRCRAFT

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      XPA = PAC
C     STORE TOTAL AVIONICS RECURRING COST FOR TEST AIRCRAFT
      AVAT = AVAC
30    IF (ZAF .EQ. 1.) GO TO 65
      IF (QT .NE. 1.) GO TO 40
C     CUMULATIVE AVERAGE TO UNIT RATIO FOR AIRFRAME LABOR
      BLA(I) = 1.0
      BMA(I) = 1.0
      GO TO 50
40    BLA(I) = (((QT+C)**0.585) - (C**0.585))/((QT**0.585) * 0.585)
C     CUMULATIVE AVERAGE TO UNIT RATIO FOR AIRFRAME MATERIAL
      BMA(I) = (((QT+C1)**0.832) - (C1**0.832))/((QT**0.832)*0.832)
C     AIRFRAME UNIT LABOR COST AT QUANTITY QT
50    ULA(I) = PAL/(QT**0.415)
C     TOTAL AIRFRAME LABOR COST FOR QUANTITY QT
      CALAC = ULA(I) * BLA(I) * QT
C     DIFFERENCE IN TOTAL AIRFRAME LABOR COST FOR QUANTITY QT AND TEST
C     AIRCRAFT
      TEMPL = CALAC - CAL
C     AIRFRAME CUMULATIVE AVERAGE LABOR COST FOR TEST AIRCRAFT
      IF (I .EQ. 1) CALA(I) = TEMPL/QT
C     AIRFRAME CUMULATIVE AVERAGE LABOR COST FOR QUANTITY QTA
      (QTA = QUANTITY QT LESS TEST AIRCRAFT)
      IF (I .EQ. 2) CALA(I) = ULA(I)
      IF (I .GT. 2) CALA(I) = TEMPL/QTA
C     AIRFRAME UNIT MATERIAL COST AT QUANTITY QT
      UMA(I) = PAM/(QT**0.168)
C     TOTAL MATERIAL COST FOR QUANTITY QT
      CAMAC = UMA(I) * BMA(I) * QT
C     DIFFERENCE IN TOTAL AIRFRAME MATERIAL COST FOR QUANTITY QT AND
C     TEST AIRCRAFT
      TEMPM = CAMAC - CAM
C     AIRFRAME CUMULATIVE AVERAGE MATERIAL COST FOR TEST AIRCRAFT
      IF (I .EQ. 1) CAMA(I) = TEMPM/QT
C     AIRFRAME CUMULATIVE AVERAGE MATERIAL COST FOR QUANTITY QTA
      IF (I .EQ. 2) CAMA(I) = UMA(I)
      IF (I .GT. 2) CAMA(I) = TEMPM/QTA
      IF (QT .LE. 0.) GO TO 60
C     CALCULATION OF AIRFRAME PRODUCTION RATE
      Q2 = QT
      IF (QT .GT. QM) Q2 = QM
      QTEMP = (3.1415927 * Q2)/(2.0 * QM)
      PR(I) = PRM * (SIN(QTEMP)**A)
      PR(I) = FLOAT(IFIX(10.0 * (PR(I) + 0.05)))/10.0
      IF (PR(I) .LT. 1.) PR(I) = 1.0
C     TOTAL AIRFRAME SUSTAINING TOOLING COST FOR QUANTITY QT
      STAC = (((PR(I)**0.4) * (QT**0.138)) - 1.0) * T1
C     DIFFERENCE IN TOTAL AIRFRAME SUSTAINING TOOLING COST FOR QUANTITY
C     QT AND TEST AIRCRAFT
      TEMPT = STAC - STAT
C     AIRFRAME CUMULATIVE AVERAGE SUSTAINING TOOLING COST FOR TEST
C     AIRCRAFT
      IF (I .EQ. 1) STA(I) = TEMPT/QT
C     AIRFRAME CUMULATIVE AVERAGE SUSTAINING TOOLING COST FOR
C     QUANTITY QTA
      IF (I .GT. 1) STA(I) = TEMPT/QTA

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      IF (QT .LE. QM) Q2 = Q2 - 1.0
      QTEMP = (3.1415927 * Q2)/(2.0 * QM)
C     PRODUCTION RATE FOR ADJUSTED QUANTITY Q2
      PR1 = PRM * (SIN(QTEMP)**A)
      PR1 = FLOAT(IFIX(10.0 * (PR1 + 0.05)))/10.0
      IF (PR1 .LT. 1.) PR1 = 1.0
C     TOTAL AIRFRAME SUSTAINING TOOLING COST FOR QUANTITY QT-1
      STA1 = (((PR1**0.4) * (Q1**0.138)) - 1.0) * TI
C     AIRFRAME UNIT SUSTAINING TOOLING COST AT QUANTITY QT
      STU(I) = STAC - STA1
C     TOTAL AIRFRAME SUSTAINING ENGINEERING COST FOR QUANTITY QT
      SEAC = EI * ((QT**0.2) - 1.0)
C     DIFFERENCE IN TOTAL AIRFRAME SUSTAINING ENGINEERING COST FOR
C     QUANTITY QT AND TEST AIRCRAFT
      TEMPE = SEAC - SEAT
C     AIRFRAME CUMULATIVE AVERAGE SUSTAINING ENGINEERING COST FOR TEST
C     AIRCRAFT
      IF (I .EQ. 1) SEA(I) = TEMPE/QT
C     AIRFRAME CUMULATIVE AVERAGE SUSTAINING ENGINEERING COST FOR
C     QUANTITY QTA
      IF (I .GT. 1) SEA(I) = TEMPE/QT
C     TOTAL AIRFRAME SUSTAINING ENGINEERING COST FOR QUANTITY QT-1
      SEA1 = EI * ((Q1**0.2) - 1.0)
C     AIRFRAME UNIT SUSTAINING ENGINEERING COST AT QUANTITY QT
      SEU(I) = SEAC - SEA1
C     TOTAL AIRFRAME CUMULATIVE AVERAGE COST FOR TEST AIRCRAFT (I = 1)
C     OR FOR QUANTITY QTA (I GREATER THAN 1)
60    AFA(I) = SEA(I) + STA(I) + CALA(I) + CAMA(I)
C     TOTAL AIRFRAME UNIT COST AT QUANTITY QT
      AFU(I) = SEU(I) + STU(I) + ULA(I) + UMA(I)
65    IF (ZEN .EQ. 1.) GO TO 200
C     QUANTITY OF ENGINES REQUIRED FOR AIRCRAFT OF QUANTITY QT
      X(I) = V * QT
      IF (I .EQ. 1) GO TO 67
      IF (I .NE. 15 .OR. INDE .EQ. 1 .OR. INDE .EQ. 3) GO TO 70
C     TOTAL ENGINE RECURRING DEVELOPMENT COST FOR TEST AIRCRAFT (I = 1)
C     OR FOR FIRST 50 PRODUCTION AIRCRAFT (I = 15)
67    IF (ENG .NE. 4.) EDAC = DEI * ((X(I)**0.07751) - 1.)
      IF (ENG .EQ. 4.) EDAC = DEI * ((X(I)**0.09334) - 1.)
C     TOTAL ENGINE PRODUCTION COST FOR AIRCRAFT QUANTITY QT
70    GO TO (80, 90, 100, 110), IENG
80    PAC = CAPE * (X(I)**0.86745)
      GO TO 120
90    PAC = CAPE * (X(I)**0.87088)
      GO TO 120
100   PAC1 = CAPE * (X(I)**0.86745)
      PAC2 = CAPE * (X(I)**0.87088)
      PAC = (TFW * PAC1) + (TFN * PAC2)
      GO TO 120
110   PAC = CAPE * (X(I)**0.89055)
C     DIFFERENCE IN TOTAL ENGINE PRODUCTION COST FOR AIRCRAFT QUANTITY
C     QT AND FOR TEST AIRCRAFT
120   TEMPP = PAC - XPA
C     ENGINE CUMULATIVE AVERAGE PRODUCTION COST FOR TEST AIRCRAFT
      IF (I .EQ. 1) PA(I) = TEMPP/X(I)
C     ENGINE CUMULATIVE AVERAGE PRODUCTION COST AT ENGINE QUANTITY VQA
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      IF (I .GT. 1) PA(I) = TEMPP/VQA
C     TOTAL OF ENGINE RECURRING DEVELOPMENT AND PRODUCTION COSTS
C     FOR TEST AIRCRAFT
      IF (I .EQ. 1) TCT = PAC
      IF (X(I) .GT. 1.) GO TO 130
      PU(I) = PA(I)
      UEA(I) = PU(I)
      GO TO 200
130   DO 190 M = 1, 2
      IF (M .EQ. 1) X1 = X(I) - 1.0
      IF (M .EQ. 2) X1 = X(I) - V
C     TOTAL ENGINE PRODUCTION COST FOR ENGINE QUANTITY X1
      GO TO (140, 150, 160, 170), IENG
140   PA1 = CAPE * (X1**0.86745)
      GO TO 180
150   PA1 = CAPE * (X1**0.87088)
      GO TO 180
160   PA1A = CAPE * (X1**0.86745)
      PA1B = CAPE * (X1**0.87088)
      PA1 = (TFW * PA1A) + (TFN * PA1B)
      GO TO 180
170   PA1 = CAPE * (X1**0.89055)
C     ENGINE UNIT PRODUCTION COST AT ENGINE QUANTITY X(I)
180   IF (M .EQ. 1) PU(I) = PAC - PA1
C     TOTAL OF ENGINE UNIT PRODUCTION COSTS PER AIRCRAFT AT AIRCRAFT
C     QUANTITY QT
      IF (V .EQ. 1.0) UEA(I) = PU(I)
      IF (V .EQ. 1.0) GO TO 200
      IF (M .EQ. 2) UEA(I) = PAC - PA1
190   CONTINUE
200   IF (ZAV .EQ. 1.) GO TO 205
C     TOTAL AVIONICS RECURRING COST FOR AIRCRAFT QUANTITY QT
      AVAC = AVI * (QT**0.81558)
C     STORE TOTAL AVIONICS RECURRING COST FOR TEST AIRCRAFT
      IF (I .EQ. 1) AVT = AVAC
C     DIFFERENCE IN TOTAL AVIONICS RECURRING COST FOR AIRCRAFT
C     QUANTITY QT AND FOR TEST AIRCRAFT
      TEMPA = AVAC - AVAT
C     AVIONICS CUMULATIVE AVERAGE RECURRING COST FOR TEST AIRCRAFT
      IF (I .EQ. 1) AVC(I) = TEMPA/QT
C     AVIONICS CUMULATIVE AVERAGE RECURRING COST FOR AIRCRAFT
C     QUANTITY QTA
      IF (I .GT. 1) AVC(I) = TEMPA/QTA
C     TOTAL AVIONICS RECURRING COST FOR AIRCRAFT QUANTITY QT-1
      IF (QT .GT. 1.) AVA1 = AVI * (Q1**0.81558)
C     AVIONICS UNIT RECURRING COST AT AIRCRAFT QUANTITY QT
      IF (QT .EQ. 1.) AVU(I) = AVC(I)
      IF (QT .GT. 1.) AVU(I) = AVAC - AVA1
C     TOTAL AIRCRAFT UNIT COST AT AIRCRAFT QUANTITY QT (AIRFRAME,
C     ENGINES, AND AVIONICS)
205   TUN(I) = AFU(I) + UEA(I) + AVU(I)
C     TOTAL CUMULATIVE AVERAGE COST FOR TEST AIRCRAFT (AIRFRAMES,
C     ENGINES, AND AVIONICS)
      IF (I .EQ. 1) TCA(I) = (TEMPL + TEMPM + TEMPT + TEMPE + TEMPP +
C     TEMPA)/QT
C     TOTAL CUMULATIVE AVERAGE COST FOR AIRCRAFT QUANTITY QTA
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C      (AIRFRAMES, ENGINES, AND AVIONICS)
      IF (I .EQ. 2) TCA(I) = TUN(I)
      IF (I .GT. 2) TCA(I) = (TEMPL + TEMPM + TEMPT + TEMPE + TEMPP +
C  TEMPA)/QTA
C  TOTAL PRODUCTION COST FOR TEST AIRCRAFT (I = 1) OR FOR AIRCRAFT
C  QUANTITY QTA (I GREATER THAN 1)
      IF (I .EQ. 1) TC(I) = TCA(I) * QT
      IF (I .GT. 1) TC(I) = TCA(I) * QTA
      IF (I .EQ. 11) DQ = 10.
      IF (I .EQ. 20) DQ = 100.
      IF (I .EQ. 29) DQ = 1000.
      IF (I .EQ. 38) DQ = 10000.
210 CONTINUE
      RETURN
      END
```

\$IBFTC OUTPUT

SUBROUTINE OUTPUT

COMMON RUN, W, S, PRM, QM, TA, ENG, T, V, RDAV, AVI, ETIN, DEI,
C CAPE, A, ZAF, ZEN, ZAV, P, CR1, CR2, CR3, CR4, CR5, CR6, TFW,
C CLEAR

COMMON AVAC, AVAT, AVA1, AVT, C, C1, CAL, CALAC, CAM, CAMAC, CT,
C CTE, DS, EDAC, EDA1, EI, FTC, PAC, PAL, PAM, PA1, QT, QTA, Q1,
C Q2, RDA, RDE, RDT, SEAC, SEAT, SEA1, STAC, STAT, STA1, TCT,
C TEMPA, TEMPE, TEMPL, TEMPM, TEMPP, TEMPT, TI, TT, VQA, XCAPE,
C XDA, XPA, X1, INDT, IENG, INDE, TFN, IRUN, IPAGE

COMMON AFA, AFU, AVC, AVU, BLA, BMA, CALA, CAMA, PA, PR, PU, Q,
C SEA, SEU, STA, STU, TC, TCA, TCOST, TUN, UEA, ULA, UMA, X, IQ,
C GDATA, IDATA, TITLE

DIMENSION AFA(40), AFU(40), AVC(40), AVU(40), BLA(40), BMA(40),
C CALA(40), CAMA(40), PA(40), PR(40), PU(40), Q(40), SEA(40),
C SEU(40), STA(40), STU(40), TC(40), TCA(40), TCOST(40), TUN(40),
C UEA(40), ULA(40), UMA(40), X(40), IQ(40), GDATA(100), IDATA(100),
C TITLE(12), F(1)

EQUIVALENCE (F(1), RUN), (T, ESH)

C
C
C

SUBROUTINE FOR PRINTING OUTPUT

1 WRITE (6, 2)
2 FORMAT (1H0, 59X, 10HINPUT DATA)
WRITE (6, 100) W, S, PRM, QM, A, TA, V
100 FORMAT (1H0//, 25X, 16HPRINCIPAL INPUTS //
C 27X, 44HGROSS TAKEOFF WEIGHT OF AIRCRAFT (IN POUNDS), 22X, F10.0/
C 27X, 44HSPEED OF AIRCRAFT (IN KNOTS) , 22X, F10.0/
C 27X, 44HMAXIMUM PRODUCTION RATE (PRM) , 22X, F10.0/
C 27X, 44HQUANTITY AT WHICH PRM IS FIRST ACHIEVED , 22X, F10.0/
C 27X, 44HEXPONENT FOR PRODUCTION RATE EQUATION , 22X, F12.2/
C 27X, 44HNUMBER OF TEST AIRCRAFT , 22X, F10.0/
C 27X, 44HNUMBER OF ENGINES PER AIRCRAFT , 22X, F10.0)
GO TO (110, 120, 130, 140), IENG
110 WRITE (6, 30) T
30 FORMAT (27X, 58HTHRUST PER ENGINE (IN POUNDS) -- TURBOJET WITH AFT
CERBURNER, 8X, F10.0)
GO TO 150
120 WRITE (6, 31) T
31 FORMAT (27X, 61HTHRUST PER ENGINE (IN POUNDS) -- TURBOJET WITH NO
C AFTERBURNER, 5X, F10.0)
GO TO 150
130 WRITE (6, 32) T
32 FORMAT (27X, 41HTHRUST PER ENGINE (IN POUNDS) -- TURBOFAN, 25X,
C F10.0)
GO TO 150
140 WRITE (6, 33) ESH
33 FORMAT (27X, 51HEQUIVALENT SHAFT HORSEPOWER PER ENGINE -- TURBOPROP
C , 15X, F10.0)
150 IF (ZEN .NE. 1.) WRITE (6, 160) TFW
160 FORMAT (27X, 46HWEIGHTING FACTOR FOR TURBOJET WITH AFTERBURNER,
C 20X, F12.2)
3 WRITE (6, 4) RDAV, AVI, P
4 FORMAT (27X, 49HAVIONICS RDTE COST (IN MILLIONS OF DOLLARS) ,
C 17X, F13.3 /
C 27X, 49HAVIONICS COST FOR UNIT 1 (IN MILLIONS OF DOLLARS), 17X,

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C F13.3 /
C 27X, 22HAIRFRAME PROFIT FACTOR, 44X, F12.2)
WRITE (6, 34) CR1, CR2, CR3, CR4
34 FORMAT (1H0/, 25X, 27HAIRFRAME ADJUSTMENT FACTORS //
C 27X, 11HENGINEERING, 55X, F12.2 /
C 27X, 11HTOOLING , 55X, F12.2 /
C 27X, 11HLABOR , 55X, F12.2 /
C 27X, 11HMATERIAL , 55X, F12.2 )
WRITE (6, 35) CR5, CR6
35 FORMAT (1H0/, 25X, 25HENGINE ADJUSTMENT FACTORS //
C 27X, 11HDEVELOPMENT, 55X, F12.2 /
C 27X, 11HPRODUCTION , 55X, F12.2 )
IF (INDE .EQ. 0) GO TO 200
WRITE (6, 36)
36 FORMAT (1H0/, 25X, 19HENGINE THRUPUT DATA /)
GO TO (170, 180, 190), INDE
170 WRITE (6, 37) DEI
37 FORMAT (27X, 41HDEVELOPMENT COST (IN MILLIONS OF DOLLARS), 25X,
C F13.3)
GO TO 200
180 WRITE (6, 38) XCAPE
38 FORMAT (27X, 60HPRODUCTION COST OF ENGINE NUMBER 1 (IN THOUSANDS OF
C DOLLARS), 6X, F13.3)
GO TO 200
190 WRITE (6, 37) DEI
WRITE (6, 38) CAPE
200 WRITE (6, 138) ENG, ETIN, ZAF, ZEN, ZAV, CLEAR
138 FORMAT (1H0/, 25X, 11HDESIGNATORS //
C 27X, 30HENGINE TYPE (ADDRESS 007) , 36X, F10.0 /
C 27X, 30HENGINE THRUPUT (ADDRESS 012) , 36X, F10.0 /
C 27X, 30HAIRFRAME DELETE (ADDRESS 016) , 36X, F10.0 /
C 27X, 30HENGINE DELETE (ADDRESS 017) , 36X, F10.0 /
C 27X, 30HAVIONICS DELETE (ADDRESS 018) , 36X, F10.0 /
C 27X, 30HCLEAR (ADDRESS 027) , 36X, F10.0 )
WRITE (6, 39) (TITLE(I), I = 1, 12), IPAGE, IRUN
39 FORMAT (1H1, 9X, 12A6, 38X, 5HPAGE , 12 / 10X, 4HRUN , 14)
IPAGE = IPAGE + 1
WRITE (6, 5)
5 FORMAT (1H0///, 36X, 58HAIRFRAME RESEARCH AND DEVELOPMENT COSTS
C (INCLUDING PROFIT) / 54X, 21H(MILLIONS OF DOLLARS) //)
WRITE (6, 6) EI, TI, DS, CT, FTC, TCT
6 FORMAT (1H0//, 11X, 2(7HINITIAL, 12X), 11HDEVELOPMENT, 12X, 4HTEST,
C 15X, 6HFLIGHT, 11X, 11HENGINES FOR / 9X, 11HENGINEERING, 10X,
C 7HTOOLING, 14X, 7HSUPPORT, 12X, 8HAIRCRAFT, 14X, 4HTEST, 14X,
C 8HTEST A/C // 7X, 6(F10.3, 10X))
WRITE (6, 7) AVT, RDA
7 FORMAT (1H0//, 8X, 12HAVIONICS FOR, 12X, 5HTOTAL /10X, 8HTEST A/C,
C 13X, 7HR AND D // 7X, 2(F10.3, 10X))
WRITE (6, 70)
70 FORMAT (1H0////////, 40X, 49HTEST AIRCRAFT PRODUCTION COSTS (INCLUD
CING PROFIT) / 54X, 21H(MILLIONS OF DOLLARS))
ITA = TA
WRITE (6, 75) ITA, SEA(1), STA(1), CALA(1), CAMA(1), AFA(1)
75 FORMAT (1H0//, 10X, 9HNUMBER OF, 10X, 2(10HSUSTAINING, 10X), 41X,
C 9HTOTAL PER / 10X, 8HTEST A/C, 11X, 11HENGINEERING, 10X,
C 7HTOOLING, 15X, 5HLABOR, 13X, 8HMATERIAL, 13X, 8HAIRFRAME // 6X,

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C I9, 12X, 5(F10.3, 10X))
  IF (ZAF .EQ. 1.) GO TO 210
  WRITE (6, 39) (TITLE(I), I = 1, 12), IPAGE, IRUN
  IPAGE = IPAGE + 1
  WRITE (6, 8)
8  FORMAT (1H0, 41X, 44HAIRFRAME PRODUCTION COSTS (INCLUDING PROFIT)/
C 53X, 21H(MILLIONS OF DOLLARS) // 45X, 36H-- COST OF TEST AIRCRAFT
C EXCLUDED --)
  WRITE (6, 9)
9  FORMAT (1H0///, 12X, 12(1X, 1H.), 10HUNIT COSTS, 12(1H., 1X), 2X,
C 8(1X, 1H.), 25HCUMULATIVE AVERAGE COSTS, 8(1H., 1X) // 13X,
C 2(10HSUSTAINING, 3X), 34X, 2(10HSUSTAINING, 3X) / 3X, 8HQUANTITY,
C 2X, 11HENGINEERING, 3X, 7HTOOLING, 6X, 5HLABOR, 6X, 8HMATERIAL,
C 5X, 5HTOTAL, 4X, 11HENGINEERING, 3X, 7HTOOLING, 6X, 5HLABOR, 6X,
C 8HMATERIAL, 5X, 5HTOTAL /)
  DO 12 I = 2, 39
  WRITE (6, 11) IQ(I), SEU(I), STU(I), ULA(I), UMA(I), AFU(I),
C SEA(I), STA(I), CALA(I), CAMA(I), AFA(I)
11  FORMAT (1H , 18, 10(2X, F10.3))
12  CONTINUE
210 IF (ZEN .EQ. 1.) GO TO 220
  WRITE (6, 39) (TITLE(I), I = 1, 12), IPAGE, IRUN
  IPAGE = IPAGE + 1
  WRITE (6, 13) RDE
13  FORMAT (1H0, 36X, 55HENGINE RESEARCH AND DEVELOPMENT COST (INCL
CUDING PROFIT) / 54X, 21H(MILLIONS OF DOLLARS) // 57X, F10.3)
  WRITE (6, 16)
16  FORMAT (1H0//, 43X, 42HENGINE PRODUCTION COSTS (INCLUDING PROFIT)/
C 53X, 21H(MILLIONS OF DOLLARS) // 40X, 48H-- COST OF ENGINES FOR
C TEST AIRCRAFT EXCLUDED --)
  WRITE (6, 17)
17  FORMAT (1H0/, 29X, 4HUNIT, 18X, 9HCUM. AVG., 38X, 10HTOTAL UNIT,
C 15X, 9HCUM. AVG. / 4X, 6HENGINE, 2(19X, 5HCOSTS), 17X,
C 8HAIRCRAFT, 18X, 5HCOSTS, 19X, 5HCOSTS / 3X, 8HQUANTITY, 16X,
C 9H(ENGINES), 15X, 9H(ENGINES), 15X, 8HQUANTITY, 15X,
C 10H(AIRCRAFT), 14X, 10H(AIRCRAFT) / )
  DO 19 I = 2, 39
  IX = V * Q(I)
  PAA = V * PA(I)
  WRITE (6, 18) IX, PU(I), PA(I), IQ(I), UEA(I), PAA
18  FORMAT (1H , 18, 1X, 2(14X, F10.3), 15X, 18, 1X, 2(14X, F10.3))
19  CONTINUE
220 IF (ZAV .EQ. 1.) GO TO 230
  WRITE (6, 39) (TITLE(I), I = 1, 12), IPAGE, IRUN
  IPAGE = IPAGE + 1
  WRITE (6, 50) RDAV
50  FORMAT (1H0, 35X, 57HAVIONICS RESEARCH AND DEVELOPMENT COST (INC
C LUDING PROFIT) / 54X, 21H(MILLIONS OF DOLLARS) // 57X, F10.3)
  WRITE (6, 52)
52  FORMAT (1H0//, 42X, 44HAVIONICS PRODUCTION COSTS (INCLUDING PRO
C FIT / 53X, 21H(MILLIONS OF DOLLARS) // 40X, 49H-- COST OF AVIONICS
C FOR TEST AIRCRAFT EXCLUDED --)
  WRITE (6, 53)
53  FORMAT (1H0/, 100X, 10HCUMULATIVE / 19X, 8HQUANTITY, 35X, 4HUNIT,
C 35X, 7HAVERAGE /)
  DO 55 I = 2, 39

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      WRITE (6, 54) IQ(I), AVU(I), AVC(I)
54  FORMAT (1H , 16X, 18, 2X, 2(30X, F10.3))
55  CONTINUE
230  WRITE (6, 39) (TITLE(I), I = 1, 12), IPAGE, IRUN
      WRITE (6, 40)
40  FORMAT (1H0, 32X, 70HCOMBINED COSTS (INCLUDING PROFIT) FOR AIRFRAM
      CES, ENGINES, AND AVIONICS / 56X, 21H(MILLIONS OF DOLLARS))
      WRITE (6, 41) RDA, RDE, RDAV, RDT
41  FORMAT (1H0/, 19X, 13HAIRFRAME RDTE / 19X, 14HINCL. TEST A/C,
      C 70X, 5HTOTAL / 20X, 12HWITH ENGINES, 11X, 6HENGINE, 13X,
      C 8HAVIONICS, 33X, 4HRDTE / 20X, 12HAND AVIONICS, 12X, 2(4HRDTE,
      C 16X), 19X, 4HCOST // 19X, 3(F10.3, 10X), 20X, F10.3)
      WRITE (6, 42)
42  FORMAT (1H0/, 43X, 4(1H., 1X), 29HPRODUCTION COSTS WITHOUT RDTE,
      C 4(1X, 1H.), 12X, 1H., 28H PRODUCTION COSTS WITH RDTE , 1H.)
      WRITE (6, 43)
43  FORMAT (1H , 20X, 10HPROD. RATE / 4X, 8HAIRCRAFT, 9X, 10H(AIRFRAMES,
      C 92X, 5HGRAND / 4X, 8HQUANTITY, 9X, 10HPER MONTH), 13X, 4HUNIT,
      C 14X, 9HCUM. AVG., 12X, 5HTOTAL, 14X, 9HCUM. AVG., 12X, 5HTOTAL /)
      DO 45 I = 2, 39
      TCQ = TCOST(I)/Q(I)
      WRITE (6, 44) IQ(I), PR(I), TUN(I), TCA(I), TC(I), TCQ, TCOST(I)
44  FORMAT (1H , 19, 7X, F10.1, 2X, 5(10X, F10.3))
45  CONTINUE
      RETURN
      END
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